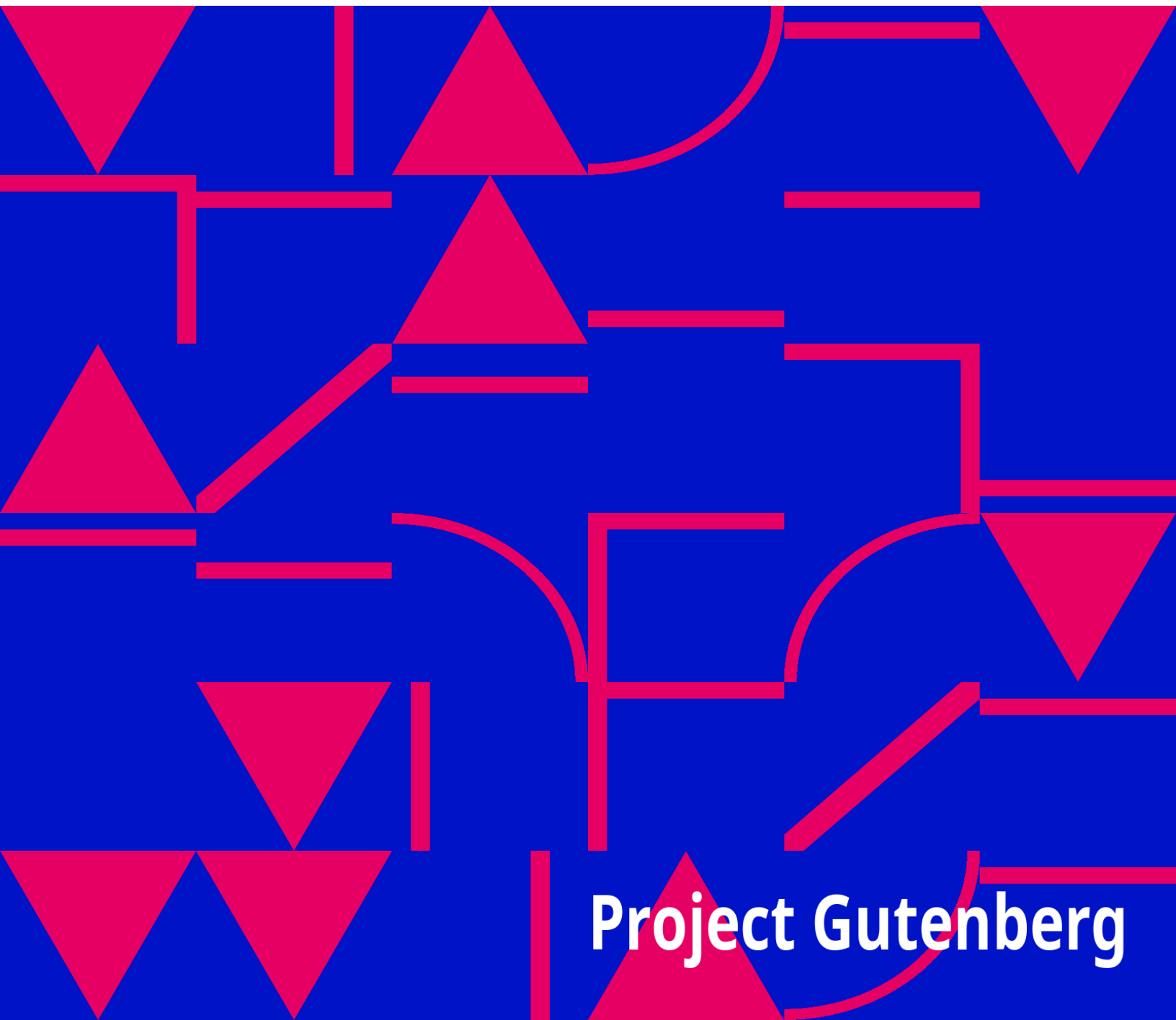


Shakespeare and Precious Stones

George Frederick Kunz

The background of the lower half of the image is a solid blue field. Overlaid on this are various red geometric shapes, including triangles of different sizes and orientations, horizontal and vertical line segments, and quarter-circle arcs. These shapes are scattered across the blue area, creating a complex, abstract pattern.

Project Gutenberg

Shakespeare and Precious Stones

George Frederick Kunz

A decorative geometric pattern on a blue background, featuring various red shapes including triangles, squares, and curved lines, creating a complex, interlocking design.

Project Gutenberg

The Project Gutenberg eBook of Shakespeare and Precious Stones

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Shakespeare and Precious Stones

Author: George Frederick Kunz

Release date: June 13, 2005 [eBook #16055]

Most recently updated: December 11, 2020

Language: English

Other information and formats: www.gutenberg.org/ebooks/16055

Credits: E-text prepared by Marilynda Fraser-Cunliffe, Sankar Viswanathan, and the Project Gutenberg Online Distributed Proofreading Team

*** START OF THE PROJECT GUTENBERG EBOOK SHAKESPEARE
AND PRECIOUS STONES ***

E-text prepared by Marilynda Fraser-Cunliffe, Sankar Viswanathan,
and the Project Gutenberg Online Distributed Proofreading Team

(<https://www.pgdp.net/>)

from page images generously provided by

the Posner Memorial Collection at <http://posner.library.cmu.edu/Posner>

Note: Images of the original pages are available through the Posner Memorial Collection. See

http://posner.library.cmu.edu/Posner/books/book.cgi?call=553_K96S_1916

SHAKESPEARE AND PRECIOUS STONES By GEORGE FREDERICK KUNZ, Ph.D., A.M., D.Sc.

THE CURIOUS LORE OF PRECIOUS STONES

Being a description of their sentiments and folklore, superstitions, symbolism, mysticism, use in protection, prevention, religion and divination, crystal gazing, birth-stones, lucky stones and talismans, astral, zodiacal, and planetary.

THE MAGIC OF JEWELS AND CHARMS

Magic jewels and electric gems; meteorites or celestial stones; stones of healing; fabulous stones, concretions and fossils; snake stones and bezoars; charms of ancient and modern times; facts and fancies about precious stones.

EACH: Profusely illustrated in color, doubletone and line. Octavo. Handsome cloth binding, gilt top, in a box. \$6.00 net. Carriage charges extra.

SHAKESPEARE AND PRECIOUS STONES

Treating of the known references to precious stones in Shakespeare's works, with comments as to the origin of his material, the knowledge of the poet concerning precious stones, and references as to where the precious stones of his time came from.

Four illustrations. Square Octavo. Decorated cloth. \$1.25 net.

MR. WILLIAM SHAKESPEARES COMEDIES, HISTORIES, & TRAGEDIES.

Published according to the True Originall Copies



LONDON

Printed by Isaac Laggard, and Ed. Blount. 1623.

WILLIAM SHAKESPEARE

Engraved by Martin Droeshout for the First Folio of 1623, wherein the
plays were first assembled.

Reproduced from a copy of this Folio owned by the New York Public

Library.
The original measures 7-1/2 x 13 in., or 20 x 33 cm.

SHAKESPEARE
and
PRECIOUS STONES

TREATING OF

The Known References of Precious Stones in Shakespeare's Works, with
Comments as to the Origin of his Material, the Knowledge of the Poet
Concerning Precious Stones, and References as to Where the Precious
Stones of his Time came from

The Author

GEORGE FREDERICK KUNZ

PH.D., Sc.D., A.M.

Honorary President of the Shakespeare Garden Committee of New York
City; Vice President of the Permanent Shakespeare Birthday Committee of
the City of New York; Member of the Executive Committee of the New
York City Tercentenary Celebration; Member of the Mayor's Shakespeare
Celebration Committee of New York.

WITH ILLUSTRATIONS



PHILADELPHIA & LONDON

IMPRINTED BY

J.B. LIPPINCOTT COMPANY

At the Washington Square Press
Upon the **TERCENTENARY** of Shakespeare

1916

PUBLISHED JULY, 1916

TO RUBY,
MY DAUGHTER,
WHOSE MOTHER, SOPHIA HANDFORTH ,
WAS BORN IN THE LAND OF
SHAKESPEARE,

AND

TO RUBY'S DAUGHTER,
GRETEL,
(THE PEARL),

THIS VOLUME IS LOVINGLY DEDICATED

FOREWORD

As no writer has made a more beautiful and telling use of precious stones in his verse than did Shakespeare, the author believed that if these references could be gathered together for comparison and for quotation, and if this were done from authentic and early editions of the great dramatist-poet's works, it would give the literary and historical student a better understanding as to what gems were used in Shakespeare's time, and in what terms he referred to them. This has been done here, and comparisons are made with the precious stones of the present time, showing what mines were known and gems were worn in Shakespeare's day, and also something of those that were not known then, but are known at this time.

The reader is also provided with a few important data serving to show what could have been the sources of the poet's knowledge regarding precious stones and whence were derived those which he may have seen or of which he may have heard. As in this period the beauty of a jewel depended as much, or more, upon the elaborate setting as upon the purity and brilliancy of the gems, the author has given some information regarding the leading goldsmith-jewellers, both English and French, of Shakespeare's age. Thus the reader will find, besides the very full references to the poet's words and clear directions as to where all the passages can be located in the First Folio of 1623, much material that will stimulate an interest in the subject and promote further independent research.

The author wishes to express his thanks to Dr. Appleton Morgan, President of the Shakespeare Society of New York; Miss H.C. Bartlett, the Shakespearean bibliophile; the New York Public Library and H.M. Leydenberg, assistant there; Gardner C. Teall; Frederic W. Erb, assistant librarian of Columbia University; the Council of the Grolier Club, Miss Ruth S. Granniss, librarian of the Club, and Vechten Waring, all of New York City.

G.F.K.

NEW YORK April, 1916

CONTENTS

PAGE

SHAKESPEARE AND PRECIOUS STONES [15](#)

PRECIOUS STONES MENTIONED IN THE PLAYS OF
SHAKESPEARE [73](#)

PRECIOUS STONES MENTIONED IN THE POEMS OF
SHAKESPEARE [93](#)

ILLUSTRATIONS

PAGE

WILLIAM SHAKESPEARE (ENGRAVED BY MARTIN DROESHOUT) [Frontispiece](#)

FIVE OF THE SIX AUTHENTIC SHAKESPEARE SIGNATURES [45](#)

DIAMOND CUTTER'S SHOP, EIGHTEENTH CENTURY [59](#)

FROM A PORTRAIT OF QUEEN ELIZABETH [61](#)

PRINTER'S MARK OF RICHARD FIELD [102](#)

SHAKESPEARE AND PRECIOUS STONES

So wide is the range of the immortal verse of Shakespeare, and so many and various are the subjects he touched upon and adorned with the magic beauty of his poetic imagery, that it will be of great interest to refer to the allusions to gems and precious stones in his plays and poems. These allusions are all given in the latter part of this volume. What can we learn from them of Shakespeare's knowledge of the source, quality, and use of these precious stones?

The great favor that pearls enjoyed in the sixteenth and seventeenth centuries is, as we see, reflected by the frequency with which he speaks of them, and the different passages reveal in several instances a knowledge of the ancient tales of their formation and principal source. Thus, in *Troilus and Cressida* (Act i, sc. 1) he writes: "Her bed is India; there she lies, a pearl"; and Pliny's tales of the pearl's origin from dew are glanced at indirectly when he says:

The liquid drops of tears that you have shed
Shall come again, transform'd to orient pearl.

Richard III, Act iv, sc. 4.

First Folio, "Histories", p. 198, col. A, line 17.

This is undoubtedly the reason for the comparison between pearls and tears, leading to the German proverb, "Perlen bedeuten Tränen" (Pearls mean tears), which was then taken to signify that pearls portended tears, instead of that they were the offspring of drops of liquid. The world-famed pearl of Cleopatra, which she drank after dissolving it, so as to win her wager with Antony that she would entertain him with a banquet costing a certain immense sum of money, is not even noticed, however, in Shakespeare's *Antony and Cleopatra*. In the poet's time pearls were not only worn as jewels, but were extensively used in embroidering rich garments and upholstery and for the adornment of harnesses. To this Shakespeare alludes in the following passages:

The intertissued robe of gold and pearl.

Henry V, Act iv, sc. 1.
First Folio, "Histories", p. 85 (page number repeated),
col. B, line 13.

Their harness studded all with gold and pearl.
Taming of the Shrew, Introd., sc. 2.
"Comedies", p. 209, col. B, line 33.

Fine linen, Turkey cushions boss'd with pearl.
Ibid., Act ii, sc. 1.
"Comedies", p. 217, col. B, line 32.

Laced with silver, set with pearls.
Much Ado About Nothing, Act iii, sc. 4.
"Comedies", p. 112, col. B, line 65.

Moreover, we have a simile which might almost make us suppose that Shakespeare knew something of the details of the pearl fisheries, when the oysters are piled up on shore and allowed to decompose, so as to render it easier to get at the pearls, for he makes one of his characters say, speaking of an honest man in a poor dwelling, that he was like a "pearl in your foul oyster". (As You Like It, Act v, sc. 4.)

In the strange transformation told of in Ariel's song, the bones of the drowned man have been turned to coral, and his eyes to pearls (Tempest, Act i, sc. 2). The strange and sometimes morbid attraction of opposites finds expression in a queer old English proverbial saying given in the Two Gentlemen of Verona: "Black men are pearls in beauteous ladies' eyes". The likeness to drops of dew appears where we read of the dew that it was "Decking with liquid pearl the bladed grass" (A Midsummer Night's Dream, Act i, sc. 1), and a little later in the same play we read the following injunction:

I must go seek some dewdrops here
And hang a pearl in every cowslip's ear.
A Midsummer Night's Dream, Act ii, sc. 1.
First Folio, "Comedies", p. 148, col. A, line 38.

And later still we have the lines:

That same dew, which sometime on the buds
Was wont to swell like round and orient pearls.
Midsummer Night's Dream, Act iv, sc. 1.
"Comedies", p. 157, col. B, line 10.

The pearl as a simile for great and transcendent value, perhaps suggested by the Pearl of Great Price of the Gospel, is used of Helen of Greece in the lines (Troilus and Cressida, Act ii, sc. 2):

She is a pearl
Whose price hath launch'd above a thousand ships.
At end of "Histories", page unnumbered
(p. 596 of facsimile), Col. A, line 19.

This being an allusion to the Greek fleet sent out under Agamemnon and Menelaus to bring back the truant wife from Troy. The idea of a supremely valuable pearl is also apparent in the lines embraced in Othello's last words before his self-immolation as an expiation of the murder of Desdemona, where he says of himself: [\[1\]](#)

Whose hand
Like the base Indian, threw a pearl away
Richer than all his tribe.
Othello, Act v, sc. 2.
"Tragedies", p. 338, col. B, line 53.

[\[1\]](#) For a Venetian tale that may have suggested these lines to Shakespeare, see the present writer's "The Magic of Jewels and Charms", Philadelphia and London, 1915, p. 393. The text of the First Folio gives "Iudean", instead of "Indian".

Although the term "Orient pearl" is that used by Shakespeare, and undoubtedly many of the older pearls of his day were really of Cinghalese or Persian origin, the principal source of supply was then the Panama fishery discovered by the Spaniards about a century earlier and actively exploited by them. [\[2\]](#) However, through the old inventories made by experts familiar with the real sources of precious stones and pearls—though not always correctly with those of the latter—the term "Orient pearl" came in time to denote one of fine hue, so that the "orient" of a pearl is still spoken of as signifying a sheen of the first quality.

[2] On the pearls brought to Europe from both North and South America in Shakespeare's time, see the writer's "Gems and Precious Stones of North America", New York, 1890, pp. 240-257; 2d. ed., 1892.

Many fine pearls of the fresh-water variety, not the marine pearls, were found in the Scotch rivers. It was these that are mentioned as having been obtained by Julius Cæsar to ornament a buckler which he dedicated to the shrine of the Temple of Venus Genetrix. It was also this type of pearl that was so eagerly sought by the late Queen Victoria when she visited Scotland. Many of these pearls exist in old, especially in ecclesiastical jewelry, and several are in the Ashburnham missal now in the J. Pierpont Morgan library.

[3] See "The Book of the Pearl", by George Frederick Kunz and Charles Hugh Stevenson, New York, 1908, colored plate opposite p. 16.

Of the glowing ruby Shakespeare seems to have known little, since he uses its name only in the conventional way to signify a bright or choice shade of red. In *Measure for Measure* (Act ii, sc. 4) the "impression of keen whips" produced ruby streaks on the skin; even more materialistic is the nose "all o'er embellished with rubies, carbuncles and sapphires" (*Comedy of Errors*, Act iii, sc. 2). The common employment of the designation carbuncle for a precious stone and also for a boil was usual from ancient times. At least, we might gather from this passage that the poet was aware of the distinction between ruby and carbuncle (pyrope garnet). Rubies as "fairy favors" is a dainty mention in the fairy drama *Midsummer Night's Dream* (Act ii, sc. 1). Cæsar's wounds "ope their ruby lips" (*Julius Cæsar*, Act iii, sc. 1). Macbeth speaks of the "natural ruby of your cheeks", in addressing his wife at the apparition of Banquo's ghost; with her this is unchanged, while with him terror or remorse has blanched it (*Macbeth*, Act iii, sc. 4). Lastly, the term "ruby lips", so often used by poets, is employed by Shakespeare with consummate art in *Cymbeline* (Act ii, sc. 2) where he writes:

But kiss; one kiss! Rubies unparagon'd,
How dearly they do't.

First Folio, "Tragedies", p. 376, col. B, line 18.

The "rubies" of the poet's time were frequently ruby spinels, or the so-called "balas rubies" from Badakshan, in Afghan Turkestan. The most noted one in the England of that period was probably the one said to have been given to

Edward the Black Prince by Pedro the Cruel of Castile, after the battle of Najera, in 1367, and now the most prized adornment of the English Crown, excepting the great historic diamond, the Koh-i-nûr. The immense Star of South Africa, weighing 531 metric carats, five times the weight of the Koh-i-nûr, is intrinsically worth much more, but lacks the manifold dramatic and historic associations of its Indian sister.

Strange to say, the beautiful sapphire is only twice named by Shakespeare, once as an adjunct to the pearl in embroidery (*Merry Wives of Windsor*, Act v, sc. 5). The single mention of chrysolite is much more impressive:

If heaven would make me such another world,
Of one entire and perfect chrysolite!

Othello, Act v, sc. 2.

"Tragedies", p. 337, col. A, line 5.

Chrysolite (peridot, or olivine) was regarded in Shakespeare's time and earlier as of exceptional rarity. The fine peridots of the Chapel of the Three Kings in Cologne Cathedral were believed to be emeralds of extraordinary size and were once valued at \$15,000,000, although they are really worth barely \$100,000; some of them are more than an inch in diameter. Whence they came is uncertain, but it is probable that they were brought from the East at some time during the Crusades. Indeed the origin of the fine peridots of the Middle Ages is shrouded in mystery; they are, however, believed to have been found in one or more of the islands in the Red Sea. In our day a number of specimens have been discovered on the small island of St. John in that sea; the deposit here is a jealously-guarded monopoly of the Egyptian Government. Peridots have also been found at Spyrges Island, in the Arabian Gulf. The most remarkable source of gem-material of this stone is meteoric, a few gems weighing as much as a carat each having been cut out of some yellowish-green peridot obtained by the writer from the meteoric iron of Glorieta Mountain, New Mexico.

That a turquoise, presumably set in a ring, was given to Shylock by Leah before their marriage, perhaps at their betrothal, is all that Shakespeare has found occasion to write of this pretty stone, one of the earliest used for adornment in the world's history, as the great mines of Nishapur, in Persia, and those of the Sinai Peninsula were worked at a very early time, the latter

by the Egyptians as far back as 4000 B.C. With the opal, the poet has seized upon its most characteristic quality, its changeableness of hue, where he says in *Twelfth Night* (Act ii, sc. 4): "Thy mind is a very opal".

A luminous ring is poetically described in one of Shakespeare's earliest plays, *Titus Andronicus*, written in or about 1590. The lines referring to the ring are highly expressive. After the murder of Bassianus, Martius searches in the depths of a dark pit for the dead body, and suddenly cries out to his companion Quintus that he has discovered the bloody corpse. As the interior of the pit is pitch dark, Quintus can scarcely believe what he hears, and he asks Martius how the latter could possibly see what he has described. The answer is given in the following lines:

Upon his bloody finger he doth wear
A precious ring, that lightens all the hole,
Which, like a taper in some monument,
Doth shine upon the dead man's earthy cheeks,
And shows the ragged entrails of the pit.

Titus Andronicus, Act ii, sc. 3.

First Folio, "Tragedies", p. 38, col. B, lines 53-57.

This certainly was suggested by the common belief in naturally luminous stones, a belief partly due to a superstitious explanation of the ruddy brilliancy of rubies and garnets as resulting from a hidden fire in the stone, and partly, perhaps, to the occasional observation of the phenomena of phosphorescence or fluorescence in certain precious stones.

It will have been seen that the text of Shakespeare's plays gives no evidence tending to show any greater familiarity with precious stones than could be gathered from the poetry of his day, and from his intercourse with classical scholars, such as Francis Bacon, Ben Jonson, and others of those who formed the unique assemblage wont to meet together at the old Mermaid Tavern in London. That a diamond could cost 2000 ducats (\$5000), a very large sum in Shakespeare's time, is noted in one of his earliest plays, the *Merchant of Venice* (Act iii, sc. 1), and the following injunction emphasizes the great value of a fine diamond:

Set this diamond safe
In golden palaces, as it becomes.

I Henry VI, Act v, sc. 3.
"Histories", p. 116, col. B, line 54.

In Pericles we read (Act iii, sc. 2):

The diamonds of a most praised water
Do appear, to make the world twice rich.

Third Folio, 1664, p. 7, col. B, line 38;
separate pagination.

In Shakespeare's time but few of the world's great diamonds were in Europe, though two, at least, were in his native country. All of them must have been of East Indian origin, as this was before the discovery of the Brazilian mines (1728). In 1547, Henry VIII of England bought of the Fuggers of Augsburg—the great money-lending bankers and jewel setters, or royal pawnbrokers, who generally sold or forced some jewels upon those who obtained a loan—the jewel of Charles the Bold, called the "Three Brethren", from three large balas-rubies with which it was set; the central ornament was a "great pointed diamond"; of its weight nothing is known. This jewel was lost by Duke Charles on the field of Granson, March 2, 1476, where it was secured by the Swiss victors; it was eventually bought by the Fuggers. The other fine English diamond was that known as the Sancy, weighing 53-3/4 carats (55.23 metric carats), acquired by James I from Nicholas Harley de Sancy, in 1604, for 500,000 crowns. This is also stated to have belonged to Charles the Bold. In 1657 it was redeemed by Cardinal Mazarin, after having been pledged for a loan by Queen Henrietta Maria, and at Mazarin's death, in 1661, was bequeathed, with his other diamonds, to the French Crown. After passing through many vicissitudes, it has recently come into the possession of Baron Astor of Hever (William Waldorf Astor).

There is a possibility that the Florentine diamond of 133-22/32 carats (137.27 metric carats) was already owned by the grand-ducal house of Tuscany before Shakespeare's death, but the earliest notice of it appears to be that given by Fermentel, a French traveller, who saw it in Florence in 1630. The other great diamonds of former days are of more recent date. The Regent of 136-7/8 carats (140.64 metric carats), found in India about 1700, was acquired by the Duke of Orleans in 1717; the Orloff (194-3/4 old carats

= 199.73 metric carats) was bought by Prince Orloff for Catherine II, in 1775, for 1,400,000 Dutch florins, or about \$560,000. The famous Koh-i-nûr, weighing 186-1/16 carats (191.1 metric carats) in its old cutting, came to Europe, as a gift to Queen Victoria from the East India Company, only in 1850; although, if it be the same as the great diamond taken by Humayun, son of Baber, at the battle of Paniput, April 21, 1526, its history dates back at least to 1304, when Sultan Ala-ed-Din took it from the Sultan of Malva, whose family had already owned it for generations.

As fresh-colored lips are likened to rubies, so it is said of a bright eye, that it "would emulate the diamond" (*Merry Wives of Windsor*, Act iii, sc. 3).

Bright eyes are also compared to rock-crystal, and the setting of other gems within a bordering of crystals is evidently alluded to in the following lines from *Love's Labour's Lost* (Act ii, sc. 1):

Methought all his senses were lock'd in his eyes
As jewels in crystal.

First Folio, "Comedies", p. 128, col. A, line 7.

We have in *Richard II* (Act i, sc. 2) the terms "fair and crystal" applied to a clear sky, and in *Romeo and Juliet* (Act i, sc. 2) the word is used to denote superlative excellence, where a lady's love is to be weighed against her rival on "crystal scales".

Rock-crystal was much more highly valued in the England of Elizabeth and of James I than it is to-day, and was freely used as an adjunct to more precious material, and still was employed to some extent in the adornment of book-covers, although this usage, so common in mediæval times, was fast passing away.

In Shakespeare's poems, "*Venus and Adonis*" (1593) and "*Lucrece*" (1594), as well as in his "*Sonnets*" (1609), in the "*Lover's Complaint*" and in the almost certainly spurious "*Passionate Pilgrim*", containing two sonnets and three poems from *Love's Labour's Lost*, and which has been included in most collections of his works, there are perhaps relatively more frequent mentions of precious stones than in the plays, a few of them being of special interest. Where we have twice "ruby lips" (and once "coral lips") in the plays, the poems speak thrice of "coral lips" or a "coral mouth"; ^[4] a

belt has "coral clasps" ("Passionate Pilgrim", l. 366). This belt bears also "amber studs", and in the "Lover's Complaint", l. 37, are "favours of amber", and also of "crystal, and of beaded jet".

[4] "Venus and Adonis", l. 542; "Lucrece", l. 420; Sonnet cxxx, l. 2.

Coming to the really precious stones, sapphire finds a single mention, also in the "Lover's Complaint", l. 215, where it is termed "heaven-hued". The same poem says of the diamond that it was "beautiful and hard" (l. 211), thus symbolizing a heartless beauty. More interesting are the following lines regarding the emerald (213, 214):

The deep-green emerald, in whose fresh regard
Weak sights their sickly radiance do amend.

This proves the poet's familiarity with the idea that gazing on an emerald benefited weak sight, an idea expressed as far back as 300 B.C. by Theophrastus, a pupil of Aristotle, and repeated by the Roman Pliny in 75 A.D. The "Lover's Complaint" furnishes another pretty line (198) contrasting the different beauties of rubies and pearls:

Of paled pearls and rubies red as blood.

In "Venus and Adonis", honey-tongued Shakespeare writes of a "ruby-colored portal".

Pearls are noted six times, usually as similes for tears, and tears are likened to "pearls in glass" ("Venus and Adonis", l. 980). A tender line is that in the "Passionate Pilgrim" (hardly from Shakespeare's hand, however):

Bright orient pearl, alack, too timely shaded.

More varied are the allusions to rock-crystal or crystal, as the poet calls it. In one place ("Venus and Adonis", l. 491) there are "crystal tears", and these form "a crystal tide" that flows down the cheeks and drops in the bosom (Idem, l. 957). On the other hand, the eyes are likened to this stone, as in "crystal eyne" ("Venus and Adonis", l. 633), or "crystal eyes" (Sonnet xlvi, l. 6). There are also "crystal favours", [5] a "crystal gate", [6] and "crystal walls", [7] the two characteristics of brilliancy and transparency suggesting these uses of the term.

[5] "Lover's Complaint", l. 37.

[6] "Idem", l. 286.

[7] "Lucrece", l. 1251.

The emeralds of Shakespeare's age had been brought from Peru by the Spaniards and had originally come from Colombian mines, such as those at Muzo, which are still worked in our day. The location of some of the early deposits here appears to have been lost sight of since the Spanish Conquest. The emeralds of Greek and Roman times, and of the Middle Ages, came from Mount Zabara (Gebel Zabara), near the Red Sea coast, east of Assuan, where traces of the old workings were found in 1817; these mines were reopened by order of Mehemet Ali, and were worked for a brief period by Mons. F. Cailliaud.

There can be no doubt that Shakespeare must have seen many fine jewels and glittering gems in pageants and processions during his residence in London. On certain special occasions the players were summoned to assist at royal functions, provision being made by the royal treasury for rich materials to be used in making special doublets and mantles for wear on these occasions. It has been suggested that the rich jewellery of many of the court portraits by Holbein and others must have impressed the poet by their wealth of color spread before his eyes; but it is nowise sure that he ever had special opportunity to closely examine such portraits, the smaller details of which may not have interested him greatly.

While it is not unlikely that some of the royal or noble ladies who attended the performances of Shakespeare's plays, while he was connected with the Globe Theatre, wore brilliant jewels, it is improbable that they were bedecked with the most valuable of their gems. The danger of being waylaid and robbed was much greater in those days than it is to-day, and it was probably only within palace or castle doors, or at some great State function, that the costliest jewels were worn. Hence nothing distantly approaching the rather excessive splendor of a New York or London opera night could ever have dazzled the poet-actor's eyes.

In the case of plays acted before the court, however, the royal and noble ladies, undoubtedly, wore many of their finest jewels, as did also the sovereign and courtiers. Still, preoccupied as Shakespeare must have been

with the presentation, or representation of the dramatic performance, he probably had little time or inclination to devote especial attention to these jewels.

No museum collections, properly so called, existed in Shakespeare's day, from which he could have acquired any closer knowledge of precious stones or gems, although the conception of a great modern museum of art and science found expression in the "New Atlantis" of his great contemporary, Lord Bacon. The modest beginnings of the Royal Society of London, founded in 1662, cannot be traced back beyond 1645. The French Academy of Sciences, founded in 1666, was preceded by earlier informal meetings of French scientists, to which allusion is even made by Lord Bacon, who died in 1626. The Berlin Academy came much later, in 1700, and the St. Petersburg Academy was first established in 1725 by Catherine I, widow of Peter the Great. One society, the Academia Secretorum Naturæ of Naples, goes back to 1560, and the Accademia dei Lincei of Prince Federico Cesi was founded at Rome in 1603. But of these Shakespeare could have known little or nothing.

That the poet knew, more or less vaguely, of America as a source of precious stones, as were the Indies, comes out in the farcical lines from *The Comedy of Errors* (Act iii, sc. 2), when one of the Dromios, in locating the various lands of the world on parts of his mistress's body, to the query of Antipholus: "Where America, the Indies?" replies: "Oh, sir, upon her nose, all o'er embellished with rubies, carbuncles, sapphires". This is the only mention of America in the plays.

A coincidence having its own significance is that April 23, the day of Shakespeare's death and also his birthday, was the day dedicated to St. George, the patron saint of Merry England. The war-cry of England is given several times by Shakespeare, as, for example:

Cry, God for Harry, England and Saint George!

Henry V, Act iii, sc. 1.

First Folio, "Histories", p. 77, col. B, line 51.

God and Saint George! Richmond and Victory!

Richard III, Act v, sc. 3.

First Folio, "Histories", p. 203, col. A, line 31.

And in I Henry VI (Act i, sc. 1) we read:

Bonfires in France forthwith I am to make,
To keep our great Saint George's feast withal.
First Folio, "Histories", p. 97, col. B, line 97.

We find no trace in Shakespeare's works of any belief in the many quaint and curious superstitions current in his day regarding the talismanic or curative virtues of precious stones. This is quite in keeping with the thoroughly sane outlook upon life that constituted the strong foundation of his incomparable mind. Not but that, like every true poet, the sense of mystery, and even the vague impression of the existence of occult powers, of the "Unknowable" in Nature, was strongly developed, but this is always in a broad and earnest spirit, far removed from all petty superstition.

Margaret of Anjou, wife of Henry VI, sacrificed her heart and diamond jewel, as a symbol of her sorrow and her love, when a tempest beat back the ship that was bearing her from the continent to the English coast. Her act, as described in the following verses, seems almost an attempt to propitiate the storm (II Henry VI, Act iii, sc. 2):

When from thy shore the tempest beat us back,
I stood upon the hatches in the storm,
And when the dusky sky began to rob
My earnest-gaping sight of thy land's view,
I took a costly jewel from my neck,
A heart it was, bound in with diamonds,
And threw it towards thy land: the sea received it,
And so I wish'd thy body might my heart.
First Folio, "Histories", p. 134, col. A, lines 41-48.

The idea of the sacredness of a ring as a love-token is voiced by Portia in Shakespeare's Merchant of Venice where she says (Act v, sc. 1):

I gave my love a ring and made him swear
Never to part with it; and here he stands;
I dare be sworn for him he would not leave it
Nor pluck it from his finger, for the wealth
That the world masters.

First Folio, "Comedies", p. 183, col. B, lines 12-16.

The nearest approach to a sentimental characterization of precious stones is to be found in "A Lover's Complaint", lines 204-217. Although we have already noted most of them separately, it may be well to give the entire passage here consecutively:

And, lo, behold these talents of their hair,
With twisted metal amorously impleach'd,
I have received from many a several fair,
Their kind acceptance weepingly beseech'd
With the annexions of fair gems enrich'd,
And deep-brain'd sonnets that did amplify
Each stone's dear nature, worth and quality.
The diamond,—why, 'twas beautiful and hard,
Whereto his invised ^[8] properties did tend;
The deep-green emerald, in whose fresh regard
Weak sights their sickly radiance do amend;
The heaven-hued sapphire and the opal blend
With objects manifold: each several stone,
With wit well blazon'd, smiled or made some moan.

^[8] Rare word, only known in this passage. Century Dictionary gives "invisible", "unseen", "uninspected", noting that some commentators suggest "inspected", "tried", "investigated".

Had Shakespeare felt much interest in the lore of gems, he had before him most of the then available material in a book of which he seems to have made some use. ^[9] This was an English rendering of the "De Proprietatibus Rerum" of Bartholomæus Anglicus (fl. ca. 1350), by Stephan Batman, or Bateman (d. 1587), an English divine and poet, who in the later years of his life was chaplain and librarian to the famous Archbishop Parker, and thus had free access to the latter's fine library. His rendering, published in 1582, bears the following quaint title: "Batman uppon Bartholome his Book De Proprietatibus Rerum"; it was published in 1582, and appears to have been widely read in England among those still interested in the learning of the scholastic period. A much earlier English version, made by John of Trevisa in 1396, was published by Wynkyn de Worde in 1495, and is considered to be the finest production of his press. ^[10]

[9] See H.R.D. Anders, "Shakespeare's Books", Berlin, 1904, pp. 238-248, and the New Shakespeare Soc. Trans., 1877-79, pp. 436 sqq.

[10] In the author's library is a fourteenth century MS. of the "De Proprietatibus Rerum", which belonged to the Carthusian Monastery of the Holy Trinity, at Dijon.

A rarely noted source for some of Shakespeare's knowledge regarding curious customs has been sought in the rambling treatise on heraldry written by Gerard Legh and issued, in 1564, under the title: "Accedens of Armorie" (approximately, Introduction to Heraldry). This is cast in the form of a dialogue between Gerard the Herehaught (Herold) and the Caligat Knight, the latter term designating an inferior kind of knight with no claim to nobility; indeed, an old writer renders it "a souldior on foot". The writer manages to weave in much material slightly or not at all connected with his main theme. Legh was the son of a Fleet Street draper. He seems to have studied a variety of subjects and gathered together many scraps of curious information. He died of the plague, October 13, 1563. His book went through several editions during Shakespeare's lifetime. Following the first edition of 1562 came successive ones in 1576, 1591, 1597, and one bearing the imprint of J. Jaggard in 1616. The author is believed to have been intentionally obscure in his treatment of heraldic questions lest he might earn the ill-will of the College of Arms by violating certain of their privileges.

While both Shakespeare and his great contemporary Cervantes died on April 23 of the year 1616, it strangely happens that Cervantes had been dead ten days when Shakespeare expired. This apparent paradox is due to the fact that while in Spain the Gregorian calendar had already been introduced, the "Old Style", or Julian reckoning, was still used in England; indeed, it was not totally abandoned until 1752, in the reign of George II, 170 years after the first use of the Gregorian reckoning on the Continent. In the seventeenth century the error to be corrected amounted to ten days, so that Shakespeare's death, under the New Style, occurred on May 3, while Cervantes died on April 13 of the Old Style.

In commemoration of the Tercentenary of Shakespeare's death, the Shakespearean scholar, Miss H.C. Bartlett, prepared for the New York Public Library an exhibition of Shakespearean books, including all the early editions of the quartos; the various editions of the folios; the works of

contemporaneous authors whom Shakespeare had consulted; and also the early works that mention Shakespeare, or cite from his plays or poems, including Greene's "Groat's Worth of Wit", published in 1592 by Henry Chettle and containing the earliest printed allusion to Shakespeare under the name of "Shake-scene".

One of the contemporary books containing citations from Shakespeare's works, shown at the New York Public Library, is "The Woman Hater", by Francis Beaumont (?1585-1615 or 1616), printed in 1607. ^[11] The citation, from Hamlet, Act i, sc. 5, ^[12] is apropos of the disappearance of a "fish head". It is put into the mouths of two of the characters, as follows:

Lazarello. Speak, I am bound to hear.

Count. So art thou to revenge when thou shalt hear.

^[11] "The Woman Hater, as it hath beene lately acted by the children of Paules, London, printed and to be sold by John Hodgers in Paules Church-yard, 1607".

^[12] First Folio, p. 257, col. B, lines 15, 16.

In the spacious hall of the beautiful Hispanic Museum in New York City there has recently been displayed, in commemoration of the tercentenary of Cervantes's death, an exceptionally fine collection of editions of his works and of rare plates illustrating episodes from them. Notable among the books was a first edition of his earliest published poems, four redondillas, a copla and an elegy, on the death, October 3, 1568, of Elizabeth de Valois, third wife of Philip II, and sister of Charles IX of France. ^[13] Dark rumors were afloat for some time that she had been poisoned by order of her husband. Among the other treasures in the Hispanic Museum exhibition was the earliest imprint of Cervantes's masterpiece, the immortal "Don Quixote". This was printed in Madrid, in 1605, by Juan de la Cuesta.

^[13] The compilation containing these poems is entitled: "Hystoria y relacio verdadera de la enfermedad felicissimo transito y sumptuosas exequias funebres de la Serenissima Reyna de España Isabel de Valoys nuestra Señora", Madrid, 1569. The opening lines of Cervantes are:

A quien yra mi doloroso canto
O en cuya oreja sonara su acento?
(To whom will my sad song go, and in
whose ears will its accents sound?)

A rather attractive bit of verse, purporting to have been written by Shakespeare and dedicated to the woman who became his wife in 1582, when he was but eighteen years old (she was eight years his senior), alludes in its third stanza to "the orient list" of gems, diamond, topaz, amethyst, emerald, and ruby. This little poem, with its play upon the lady-love's name, can find a place here, although many readers are already familiar with it.

TO THE IDOL OF MINE EYES AND THE DELIGHT
OF MINE

HEART,

ANNE HATHAWAY

Would ye be taught, ye feathered throng,
With love's sweet notes to grace your song,
To pierce the heart with thrilling lay,
Listen to mine Anne Hathaway!
She hath a way to sing so clear,
Phœbus might wond'ring stop to hear;
To melt the sad, make blithe the gay,
And nature charm, Anne hath a way:
 She hath a way,
 Anne Hathaway,
To breathe delight Anne hath a way.

When envy's breath and rancorous tooth
Do soil and bite fair worth and truth,
And merit to distress betray,
To soothe the heart Anne hath a way;
She hath a way to chase despair,
To heal all grief, to cure all care,
Turn foulest night to fairest day:
Thou know'st, fond heart, Anne hath a way,
 She hath a way,
 Anne Hathaway,
To make grief bliss Anne hath a way.

Talk not of gems, the orient list,
The diamond, topaz, amethyst,

The emerald mild, the ruby gay;
Talk of my gem, Anne Hathaway!
She hath a way, with her bright eye,
Their various lustre to defy,
The jewel she and the foil they,
So sweet to look Anne hath a way.
 She hath a way,
 Anne Hathaway,
To make grief bliss Anne hath a way.

But were it to my fancy given
To rate her charms, I'd call them Heaven;
For though a mortal made of clay,
Angels must love Anne Hathaway.
She hath a way so to control
To rupture the imprisoned soul,
And sweetest Heaven on earth display,
That to be Heaven Anne hath a way!
 She hath a way,
 Anne Hathaway,
To be Heaven's self Anne hath a way.

This little poem is by Charles Dibdin (1748-1814), the writer of about 1200 sea-songs, at one time great favorites with sailors. It appeared, in 1792, in his long-forgotten novel, "Hannah Hewit, or the Female Crusoe", and Sir Sidney Lee conjectures that it may have been composed on the occasion of the Stratford jubilee of 1769, in the organization of which Dibdin aided the great actor, David Garrick. In the "Poems of Places", New York, 1877, edited by Henry W. Longfellow, this poem is assigned to Shakespeare on the strength of a persistent popular error. ^[14] In his "Life" Dibdin says: "My songs have been the solace of sailors in their long voyages, in storms, in battle; and they have been quoted in mutinies to the restoration of order and discipline". It has been asserted that they brought more men into the navy than all the press gangs could do.

^[14] Sir Sidney Lee, "A Life of Shakespeare", new edition, London, 1915, p. 26, note.

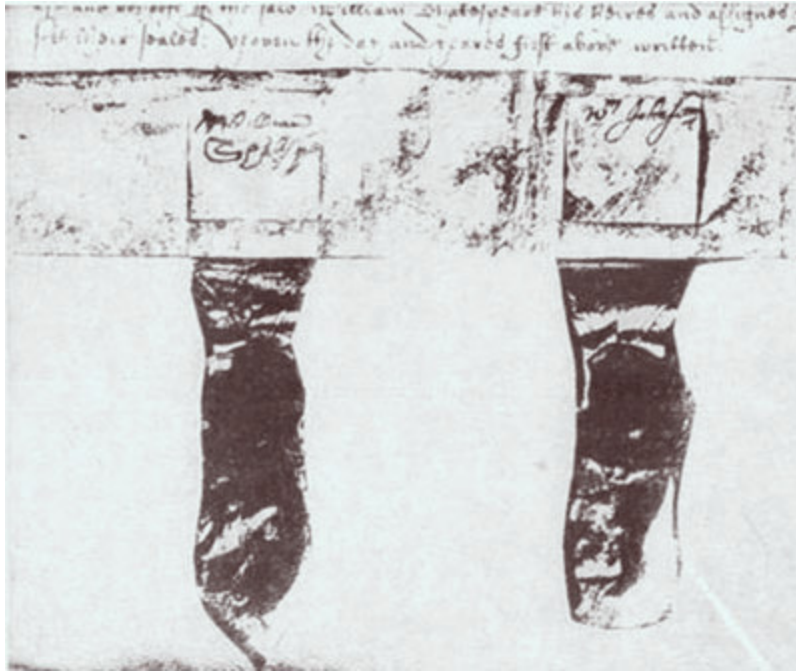
The poem has sometimes been attributed to Edmund Falconer (1814-1879), an actor and dramatist, born in Dublin, and whose real name was Edmund O'Rourke. However, his poem entitled "Anne Hathaway, A Traditionary Ballad sung to a Day Dreamer by the Mummers of Shottery Brook", ^[15] falls far below the lines we have quoted in poetic quality, as may be seen from the opening stanza (the best), which runs as follows:

No beard on thy chin, but a fire in thine eye,
With lustiest Manhood's in passion to vie,
A stripling in form, with a tongue that can make
The oldest folks listen, maids sweethearts forsake,
Hie over the fields at the first blush of May,
And give thy boy's heart unto Anne Hathaway.

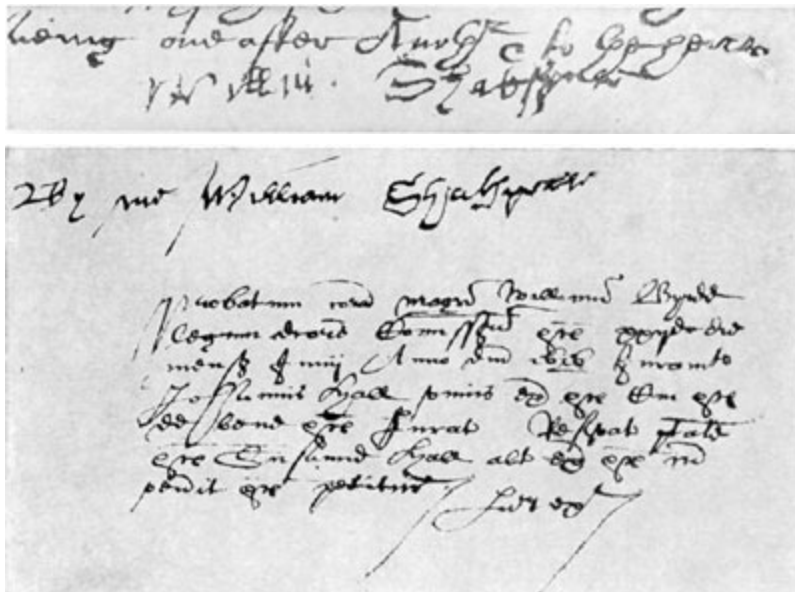
^[15] Edmund Falconer, "Memories, the Bequest of my Boyhood", London, 1863, pp. 14-22.

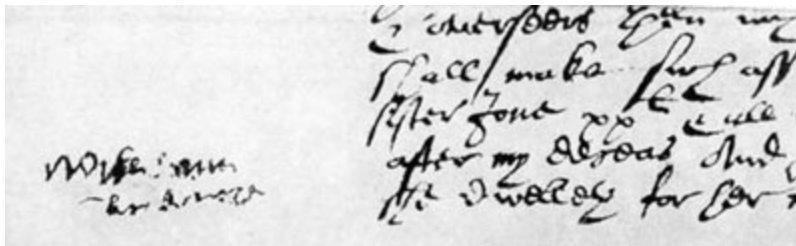
In none of the allusions to precious stones made by Shakespeare is there any indication that he had in mind any of the Biblical passages treating of gems. The most notable of these are the enumeration of the twelve stones in Aaron's breast-plate (Exodus xxviii, 17-20; xxxix, 10-13), the list of the foundation stones and gates of the New Jerusalem given by John in Revelation (xxi, 19-21), and the description of the Tyrian king's "covering" in Ezekiel (xxviii, 130). Had the poet given any particular attention to these texts we could scarcely fail to note the fact. Other Bible mentions, such as those elsewhere made by Ezekiel (xxvii, 16, 22), regarding the trade of Tyre, the agates (and coral) from Syria, and the precious stones brought by the Arabian or Syrian merchants of Sheba and Raamah, are too much generalized to invite any special notice. The same may be said of most of the remaining brief allusions. We might rather expect that where the color or brilliancy of a precious stone is used as a simile this might strike a poet's fancy and perhaps find direct expression in his own words. The light of the New Jerusalem is likened to "a jasper stone, clear as crystal" (Rev. xxi, 11), and in Exodus (xxiv, 10) the sapphire stone is said to be "as it were the body of heaven in its clearness". However, that Shakespeare wrote of "the heaven-hued sapphire" ("Lover's Complaint", l. 215) has no necessary connection with this, as the celestial hue of the beautiful sapphire is spoken of time and again by many of the older writers.

FIVE OF THE SIX AUTHENTIC SHAKESPEARE SIGNATURES

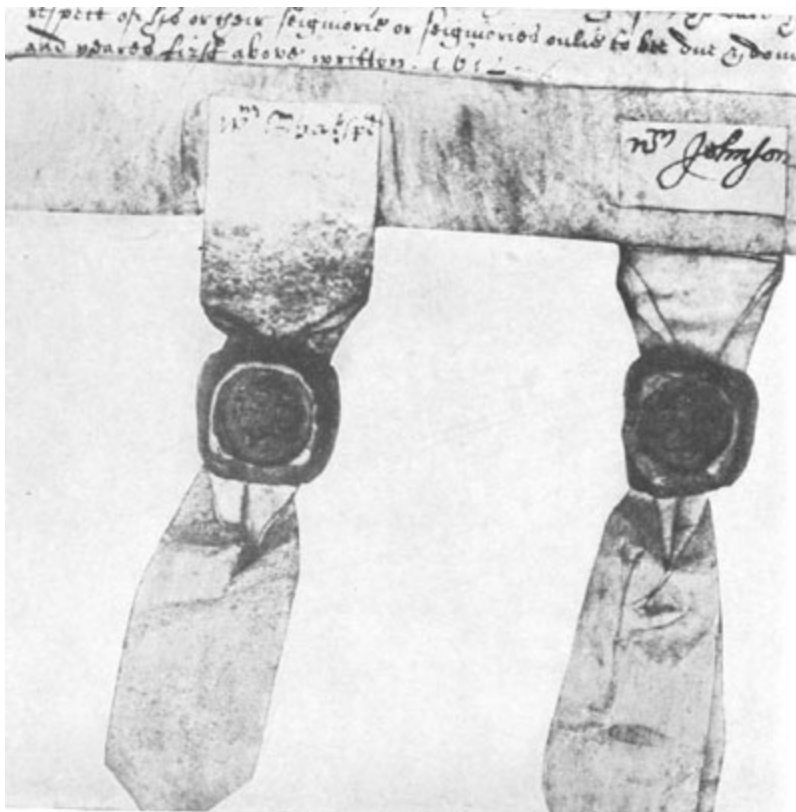


Signature on the purchase deed of Shakespeare's house in Blackfriars dated March 10 1613. In the Guildhall, London.





Signatures on the three pages of Shakespeare's will executed March 25, 1616. Original in Somerset House, London.



Signature attached to the deed mortgaging the house in Blackfriars, dated March 11, 1613. In the British Museum.

It should be borne in mind that the great English translation of the Bible, popularly called "King James' Bible", was published only after Shakespeare had completed his last play in 1611. Before that time, dating from Tyndale's version of 1525, and in great measure based on it, a number of English translations had appeared, the most authoritative in Shakspeare's time being

perhaps the "Bishops' Bible", printed under the patronage of Queen Elizabeth in 1568, and edited by the Archbishop of Canterbury.

The Geneva Bible of 1560, the first entire Bible in English in which the division into chapters and verses was carried out, had, however, the widest dissemination in Shakespeare's time, and a careful study of passages in his works referable to Biblical texts appears to prove that this version was the one with which he was most familiar. His plays testify to his close knowledge of the Scriptures, although no writer is less fettered by purely doctrinal considerations. The Geneva Bible went through no less than sixty editions in Queen Elizabeth's reign, and even after the issue of the "Authorized Version" in 1611 it competed successfully with this for a time.

That Shakespeare may have seen Philemon Holland's (1552-1637) excellent translation of Pliny is nowise unlikely. A notable passage in his *Othello* seems in any case to indicate that it was suggested by Pliny's words (Bk. II, chap. 97, in Holland's version):

And the sea Pontus evermore floweth and runneth out into Propontic, but the sea never retireth backe againe within Pontus.

Othello replies thus to Iago's conjecture that he may change his mind (Act iii, sc. 3):

Never, Iago. Like to the Pontic sea,
Whose icy current and compulsive course
Ne'er feels retiring ebb, but keeps due on
To the Propontic and the Hellespont,
Even so my bloody thoughts, with violent pace,
Shall ne'er look back, ne'er ebb to humble love.

First Folio, "Tragedies", p. 326, col. B, lines 34-39.

There is, however, no trace of any familiarity on Shakespeare's part with the precious stone lore of the Roman encyclopædist, either from the Latin text of his great "*Historia Naturalis*", or from the translation published by Holland in 1601. This translator, who Englished many of the chief Latin and Greek authors, Suetonius, Livy, Ammianus Marcellinus, Plutarch's "*Morals*" and other works, was pronounced by Fuller, in his "*Worthies*", to be "translator general in his age", adding that "these books alone of his

turning into English will make a country gentleman a competent library". For his Ammianus Marcellinus the Council of Coventry, his place of residence, paid him £4, and £5 for a translation of Camden's "Britannia"—small sums, indeed, for so much labor, but not so unreasonable when we think that a half-century later the immortal Milton got but £5 for his "Paradise Lost". He was a Fellow of Trinity College, Cambridge, where he had studied and graduated; later he studied medicine, receiving a degree of M.D., not from Oxford or Cambridge, however, but either from a Scottish or foreign university.

Although Solinus, writing in the third century A.D., relies mainly upon Pliny for his information on precious stones, still he here and there gives evidence of a more critical spirit, as when he says of the rock-crystal that the theory according to which it was frozen and hardened water was necessarily incorrect, for it was to be found in such mild climates as "Alabanda in Asia and the island of Cyprus". ^[16] This is the more notable that the wholly incorrect view persisted into the sixteenth century, so learned a writer as Lord Bacon (d. 1626) restating it in his last work, "Sylva Sylvarum".

^[16] Collectanea rerum memorabilium, Cap. 15.

One of the most curious gem-treatises, especially as a source of early sixteenth-century beliefs in the magic properties of precious stones, the "Speculum Lapidum" of Camillo Leonardo, published in Venice, 1502, probably never came under Shakespeare's eye. Indeed, even in Italy it seems to have been so neglected that Ludovico Dolci ventured to publish a literal Italian version of the Latin original as his own work in 1565. The English "Mirror of Stones", issued in 1750, is frankly stated to be a translation of the Latin original bearing the same name. ^[17]

^[17] Noted in the present writer's "The Curious Lore of Precious Stones", Philadelphia and London, 1913, p. 18.

In Marlowe's (1564-1593) "Hero and Leander", almost certainly written before Shakespeare's "Venus and Adonis" (1593), although not published until 1598, five years after Marlowe's death, "pearl tears" and the "sparkling diamond" are used much in the same way as by Shakespeare, as appears in the following verses:

Forth from those two translucent cisterns brake
A stream of liquid pearl, which down her face
Made milk-white paths.

Lines 296-298.

Why should you worship her! her you surpass
As much as sparkling diamonds flaring glass.

Lines 213,214.

There is a curious parallelism between a passage in *Troilus and Cressida*, 1609, and one in Marlowe's *Dr. Faustus*, 1588. Marlowe wrote (sc. 14, l. 83):

Was this the face that launched a thousand ships
And burnt the topless towers of Ilium?

This is followed very closely by Shakespeare, with the substitution of "pearl" for "face".

She [Helen] is a pearl,
Whose price hath launch'd above a thousand ships.
Troilus and Cressida, Act ii, sc. 2, l. 82.
First Folio, at end of "Histories", unnumbered page
(596 of facsimile), col. A, line 19.

The greatest of the world's poets lived in a period midway between the highest development of Renaissance civilization and the foundation of our modern civilization, and he was thus at once heir to the rich treasures of a glorious past, and endowed with a poetic, or we might say a prophetic insight that makes his works appeal as closely to the readers of to-day as to those of his own time.

In the four leading European nations of the age—Italy, despite her high rank in art, still lacked national unity—four sovereigns of marked though widely diverse character and attainments reigned for a considerable part of Shakespeare's life. Of the "Virgin Queen" we scarcely need to write. The England of her day, and of later days, would not have been what it was and what it became, without the aid of her mingled shrewdness and prudence. Faults she had and shortcomings, but, granted the almost overpowering difficulties she had to face, both at home and abroad, it is doubtful whether

a more decided, a more straight-forward policy would have been as successful as the somewhat devious one she pursued. Her chief rival, Philip II (1556-1598), as much averse as Elizabeth herself to energetic action, even more fond of procrastination, lacked her relative religious and political tolerance, and left Spain weaker than he had found it. And still his tenacity, his devotion to the cause he believed to be that of heaven, his consistency, and even the gloomy seriousness of his life, testify to a strong soul, though a thoroughly unlovable one.

The reign of the eccentric Rudolph II, Emperor of Germany (1576-1612), whose imperial residence was at Prague, covers the greater part of Shakespeare's life. In spite of many failings and mistakes, this monarch did much to foster the study of the arts and sciences of his age, so far as he was able to understand them. That he was for a time the dupe of adventurers and alchemists, such as the half-visionary John Dee and the altogether unscrupulous Edward Kelley, was no unusual experience in those days, when the dividing line between true science and charlatanism was too indistinctly marked to be easily discernible.

The greatest of all the sovereigns of Shakespeare's time was Henry IV of France, unquestionably the greatest of French kings, despite the fact that the primacy has often been accorded to the Roi Soleil, Louis XIV. The powerful and ductile personality that was able to put an end to the destructive religious wars of France and to lay a firm foundation for the strongly-centralized power of a later time, a foundation which the great statesman Richelieu broadened and deepened, deserves all the credit that should be given to those who conquer the first apparently insurmountable difficulties in the realization of a great aim.

How brief was the reign of most of the popes of this time is shown by the fact that no less than ten of them were at one time or other Shakespeare's contemporaries, although the duration of his life was but fifty-two years. Of these probably the most noteworthy was Gregory XIII (1572-1585), in whose reign occurred the fearful Massacre of St. Bartholomew, August 24, 1572, and the reform of the calendar from that known as the Julian to the new style named the Gregorian Calendar in honor of this pope.

In the East, just coming into closer commercial intercourse with Europe, the long reign of the greatest of the Mogul emperors, Jelal-ed-din Akbar (1556-1605), began two years before the accession of Elizabeth and lasted two years after her death. Probably no Oriental sovereign, certainly no Indian sovereign, ranks higher than Akbar, who was at once a great statesman, an able organizer, and singularly tolerant in religion. In Persia, one of the most marked rulers of this land, Abbas the Great, began to reign in 1584 and died in 1628.

In no period was jewelry worn more ornately, or with greater display, we might almost say ostentation, than in the age of Shakespeare. As a rule, in this period the precious stones were less considered than the elaborate goldsmith work in which they were placed. They were the adjuncts, rather than the principal glory of the jewel.

The court jeweller of James VI of Scotland and of this monarch after his accession to the English throne, as James I, was George Heriot (ca. 1563-1624), born in Edinburgh, the son of a member of the company of goldsmiths in that city. As the Scotch goldsmiths cumulated the profession of money-lending with that of goldsmithing, they were usually persons of considerable account among the citizens. Heriot became a member of the company in 1588, the year of the Spanish Armada. Despite the rather straitened circumstances of the Scottish court, considerable amounts were expended for jewels, especially as the queen, Anne of Denmark, was very fond of display. The nobility also, such of them at least as possessed the means, were inclined to deck themselves out with brilliant jewels and splendid ornaments of massive gold. Heriot's appointment as goldsmith to the queen dates from 1597; soon after this he was made jeweller and goldsmith to the king. He followed the court to London in 1603, when King James succeeded to Elizabeth, and at the time of his death, February 12, 1624, had amassed the sum of £50,000 by his profitable connection with the court, and had also acquired lands and houses at Rochampton, in Surrey, and St. Martin's-in-the-Fields, London. His residuary estate, which amounted to £23,625 (\$118,125), he entrusted to the provosts, bailiffs, ministers, and ordinary town-council of Edinburgh for the erection of an institution to be called Heriot's Hospital, where a number of poor freemen's sons of the town should be educated. ^[18] This foundation still exists, and the excellent management of those who have had to do with the endowment is

shown by the fact that the income it now produces equals the whole sum of the original bequest.

[18] William Hone, "The Every-Day Book", London, 1838, vol. ii, cols. 748, 749.

This great Scotch goldsmith fashioned a number of splendid rings for the queen. An old account furnished by Heriot lists them as follows: [19]

A ring with a heart and serpent, all set about with diamondis;

A ring with a single diamond, set in a heart betwixt two hands;

A great ring in the form of a perssed hand and a perssed eye, all sett with diamondis;

One great ring, in forme of a frog, all set with diamondis, price two-hundreth poundis;

A ring of a burning heart set with diamondis;

A ring in the forme af a scallope shell, set with a table diamond, and opening on the head;

A ring of a love trophe set with diamondis;

Two rings, lyke black flowers, with a table diamond in each;

A daissie ring sett with a table diamond;

A ryng sett all over with diamondis, made in fashion of a lizard, 120 l.;

A ring set with 9 diamondis, and opening on the head with the King's picture in that.

[19] William Hone, "Every-Day Book", London, 1838, vol. ii, cols. 749, 750.

Heriot also lists a ring delivered about 1607 to Margaret Hartsyde, one of the royal household, describing it as "sett all about with diamondis, and a table diamond on the head"; that is, in the bezel. He states that he had been given to understand that this was by direction of Her Majesty. His precaution in making this note appears to have been fully justified, for this Margaret Hartsyde was tried in Edinburgh, May 31, 1608, on the charge of

having purloined a pearl belonging to the queen and valued at £110. Her excuse was that she had taken this and other pearls to adorn dolls for the amusement of the royal children, and that she did not expect the queen would ask for them. As, however, it was brought out in the trial that she had cleverly disguised some of the pearls she had taken, and had offered to sell them to the queen, she was condemned to imprisonment in Blackness Castle until the payment of a fine of £400, and to confinement in Orkney during the remainder of her life. Eleven years later, however, the king's advocate "produced a letter of rehabilitation and restitution of Margaret Hartsyde to her fame". ^[20]

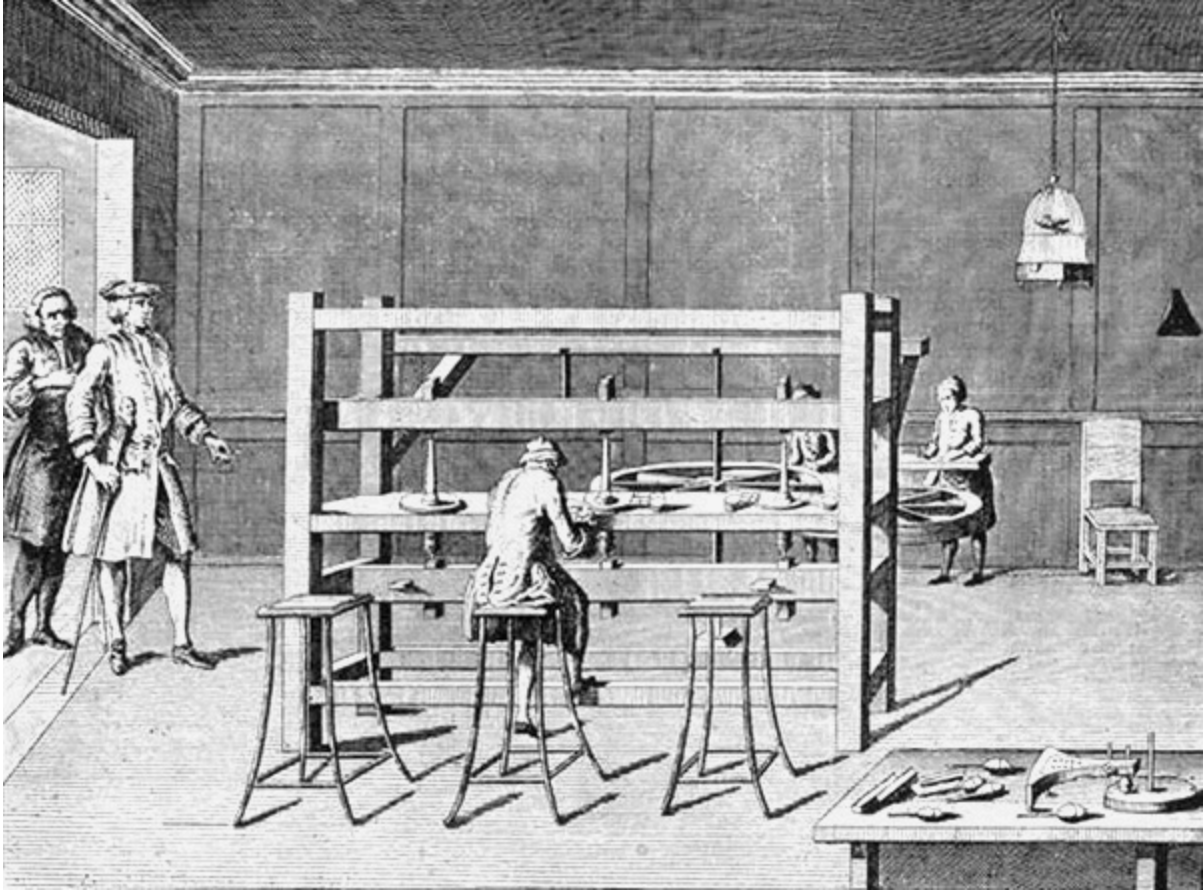
^[20] "Every-Day Book", loc. cit.

In Shakespeare's day the "goldsmiths" were also jewellers and gem dealers, and often money-lenders as well. The settings of the finest precious stones were at that time generally of gold, rarely of silver. Platinum, the metal that now enjoys the greatest furore for diamond settings, was then unknown in Europe; it was first brought to Europe in 1735, from South America, having been found in the alluvial deposits of the river Pinto, in the district of Choco, now forming part of the United States of Colombia. The Spaniards had named it platina, from its resemblance to plata, silver. The chief source in our time is Russia, the richest deposits being those discovered in 1825, on the Iss, a tributary of the Tura, in the Urals. Other valuable deposits are in the district of Nizhni-Tagilsk. Platinum also occurs in Brazil, California, and British Columbia, associated with gold, as well as in Borneo, New South Wales, Australia, and in New Zealand. Its use in gem-mountings began about 1870, and from 1880 onward it has become more and more favored, until now it has almost entirely superseded gold in the finest jewelry, especially for diamond settings. Long before the metal was known and used in Europe, ornamental use of it was made in South America, in the district we have mentioned, the material not being fused, but simply forged out of the nuggets found in the deposits.

That but few fine diamonds were in Europe when Shakespeare wrote has already been noted; indeed, the annual importation from India, then the only source, can hardly have exceeded \$100,000 on an average, while at the present day the value of the diamonds from the great African mines

imported into Europe and America amounts to from \$40,000,000 to \$60,000,000 each year.

In King James's reign, besides Heriot, William Herrick (brother of Nicolas) and John Spilman were appointed jewellers to the king, queen, and prince, the annual emoluments being £50 annually. It is stated that Herrick furnished jewels worth £36,000 to Queen Anne of Denmark. Such of her many jewels as were to be found when she died are said to have been left to her son, later Charles I, and none to her daughter Elizabeth, later Queen of Bohemia and ancestress of many of the sovereigns of Europe, as well as of the present reigning house in England. Unfortunately for her heir, a great part of the jewels had been embezzled, and could not be recovered, although models of many had been carefully preserved by William Herrick, who swore that the originals had been delivered to the queen. Less notable jewellers of King James's day were Philip Jacobson, Arnold Lulls, John Acton, and John Williams. One of them, Arnold Lulls, has left a fine set of contemporary drawings representing jewels of the epoch; these are now to be seen in the Victoria and Albert Museum, London. As an instance of the value of some of the jewels of his design, it is recorded that the sum of £1550 was paid for a diamond jewel with pearl pendants and two dozen buttons, furnished to the king to be bestowed upon the queen at the christening of the Princess Mary in 1605. [\[21\]](#)



Diamond cutter's shop, eighteenth century, in which the diamond-cutting mill is operated by "man-power". Published in the Universal Magazine of Knowledge and Pleasure, by John Hinton, England, July, 1749

[21] H. Clifford Smith, "Jewellery", London, 1908, p. 302.

While the jeweller's art in England was still under the influence of foreign goldsmiths in Elizabeth's time, it had to a considerable extent emancipated itself from foreign control in the latter part of her reign and in that of her successor. In addition to George Heriot, whom we have just noticed, several others are well worthy of mention, such as Dericke Anthony, Affabel Partridge, Peter Trender, and Nicolas Herrick, [22] the father of the poet Robert Herrick, who makes many a telling use of the colors and charm of precious stones and pearls in his dainty poems. To these must be added Sir John Spilman, of German birth, who made many jewels at the royal command.

[22] H. Clifford Smith, "Jewellery", London, 1908, pp. 219, 220, 301.

We should remember that for the cutting of precious stones steam-power was not then available, "man-power" being employed. A large turning wheel was pushed around by a man holding a bar extending from it. The motion of this large wheel was transmitted to other smaller ones. The number of revolutions per minute hardly exceeded a few hundred, while in modern times a speed of from 2000 to 2500 revolutions per minute is attained. The diamond cutting industry was largely in the hands of Jews in Lisbon.

The gem-cutting processes were not greatly modified for many years after Shakespeare's death, so that a representation of the wheel and mill used in 1750 gives a fairly good general idea of the *modus operandi*. The large wooden wheel, whose axis is the second pillar within the frame, is bent, and makes an elbow under the wheel to receive the impulsion of a bar that serves instead of a turn-handle. On the right side of the frame, where the boy stands, is the turn-handle which sets the wheel in motion by means of the elbow of its axis. So that if the wooden wheel be twenty times larger than the iron one, a hundred turns of the larger wheel will cause a thousand revolutions of the smaller one. The method of holding the diamond in place over the iron wheel, when in motion, so that it presses upon the latter and is polished thereby, is shown in the lower right-hand corner of the plate.

The German traveller, Paul Hentzner, who visited England in 1598, toward the end of Elizabeth's life, describes her jewelling in the following words:

"The Queen had in her ears two pearls with very rich drops; she wore false hair and that red; upon her head she had a small crown; her bosom was uncovered, and she had on a necklace of exceedingly fine jewels. She was dressed in white silk, bordered with pearls of the size of beans, and over it a mantle of black silk shot with silver threads; her train was very long. Instead of a chain, she had an oblong collar of gold and jewels".



FROM A PORTRAIT OF QUEEN ELIZABETH In the possession of his Grace the Duke of Devonshire, K.G., Hardwick Hall. The queen has jewels in her hair, a pearl eardrop, and two necklaces, one fitting closely to the neck, the other falling over the breast. The stiff brocade skirt is embroidered with a wonderful array of aquatic birds and animals. On the left, the cushion of the chair of state is embroidered with the queen's monogram. Surmounting the chair is a crystal ball.
The original canvas measures 90x66 inches.

In addition to this display the traveller tells us that the queen's right hand was fairly sparkling with jewelled rings.

Aside from his portrayal of jewels in his numerous portraits, Holbein ranked as the master designer of jewels in his day. Many of the finest of these designs have been preserved for us and can be seen in the British Museum, to which they were bequeathed by Sir Hans Sloane in 1753. There are 179 separate pieces, usually pen-and-ink sketches. The execution of the jewels from these designs is believed to have been mainly done by Hans of Antwerp, known as Hans Anwarpe, a friend of Holbein, who settled in London in 1514, and was appointed goldsmith to King Henry VIII, for whom he produced many jewels for New Year's gifts. ^[23]

^[23] H. Clifford Smith, "Jewellery", London [1908], pp. 211, 213.

In judging of the jewels figured in portraits we must remember that the artist has often modified them to bring them into greater harmony with their immediate surroundings. This, in some cases, may lead him to make of a somewhat inartistically designed jewel a beautifully proportioned one. Again, he may be led to exaggerate the size of the precious stones or pearls, and to intensify or deepen their colors. A recent instance regards a portrait of the former queen of Spain by one of the foremost Spanish artists of our day. The royal lady was depicted wearing an enormous pearl; however, the artist informed the author that the real pearl was much smaller than the painted one, but that, in portraying it, a better decorative effect was obtained by increasing its size. Whether Holbein (1497-1543), with his Dutch exactness of portrayal, was led into any similar exaggerations we can never tell, as little as we can know anything definite regarding the true size of the jewels shown in the portraits by the Italian Zuccherro (1529-1566), the Fleming Lucas de Heere (1524-1584), or by any other of the portrait painters of Elizabeth's time.

In a very modest way the addition of gilded scarf-pins, brooches, chains, etc., not owned by the sitters, was not uncommonly practised thirty or forty years ago, when colored tintypes were popular. These were painted on the photographs, much to the gratification of those who ordered them for distribution among their friends.

The court-jewellers of France in Shakespeare's day rivalled, though they did not excel, those of England. Among them a prominent place belongs to Francois Dujardin (or Desjardin), goldsmith of Charles IX (1560-1574) and Henri III (1574-1589). When a verification and an inventory of the French

Crown Jewels were made on August 1, 1574, after the death of Charles IX, the expert examination was entrusted to François Dujardin, who is termed "orfevre et lapidaire du Roy". The goldsmith's art was passed down from father to son in this family: a second F. Dujardin (b. ca. 1565) mounted the parures made for Elizabeth of Austria, daughter of Henri IV and Maria de' Medici. In the reign of Henri IV and the succeeding regency of Maria de' Medici, Josse de Langerac, received as master goldsmith in 1594, and the brothers Rogier, are noted as leading goldsmiths who, besides executing many fine jewels, frequently made loans of money to the Queen Regent, and seem to have experienced great difficulty in securing full payment. Corneille Rogier set the jewels worn at her marriage by Anne d'Autriche, wife of Louis XIII. Two brothers, each bearing the name Pierre Courtois, are also noted in old records. One of them, at the time of his death, in 1611, occupied two apartments with two shops in the Louvre; the shop of the other had the sign "Aux Trois Roys", probably referring to the "Three Kings of the East", the Magi of the Gospel, very appropriate patrons for goldsmiths. ^[24]

^[24] Germain Bapst, "Histoire des Joyaux de la Couronne de France", Paris, 1889, pp. 175, 176, 300, 304.

Thierry Badouer, a German goldsmith-jeweller, received from the French court, in 1572, an order for 250,000 crowns' worth of jewels to be distributed as gifts at the approaching marriage of Henri de Navarre with Marguerite de Valois. He faithfully executed his part of the task and brought the jewels with him to Paris, but before he had been able to deliver them to the Royal Treasury they were stolen from him during the confusion of the St. Bartholomew Massacre. Eventually, in the reign of Henri IV, his widow was partly reimbursed for the loss, receiving one-quarter of the amount of her claim. ^[25] After the Massacre of St. Bartholomew, and as a result of it, many Protestants and Catholics left France for Hanau, Germany, where to this day they carry on the jeweller's art; and from this beginning Hanau became a jeweller's centre.

^[25] Op. cit., p. 289.

The best reproduction of the First Folio of 1623 is the photographic facsimile, made in 1902, of the copy formerly owned by the Duke of Devonshire and now in the possession of Henry E. Huntington, of New

York.^[26] The original Folio, prepared by the managers of Shakespeare's company, John Heminge and Henry Condell, bears the imprint of Isaac Jaggard and Edward Blount, the printing house being conducted by William Jaggard and his son Isaac. It is believed that an edition of five hundred copies was issued, at one pound per copy. That the publication was essentially a commercial venture, although it may also have been a labor of love for some of the editors, is brought out clearly and quaintly in the preface addressed to "The great Variety of Readers", and signed by Heminge and Condell. This reads that the book was printed at the charges of W. Jaggard, Ed. Blount, I. Southweeke, and W. Apsley, 1623. The following passage from the preface is well worth quoting, its spirit is so delightfully modern:

The fate of all Bookes depends upon your capacities, and not of your heads alone, but of your purses. Well! It is now publique, & you wil stand for your priviledges, wee know: to read, and censure.^[27] Do so, but buy it first. That doth best commend a Booke the Stationer sales. Then, how odde soever your braines be, or your wisdomes, make your license the same and spare not.... But whatever you do, Buy. Censure will not drive a Trade, nor make the Jacke go.

^[26] "Shakespeares Comedies, Histories, & Tragedies, being a reproduction in facsimile of the First Folio Edition of 1623, from the Chatsworth copy in the possession of the Duke of Devonshire, K.G., with introduction and censure of copies by Sidney Lee". Oxford, Clarendon Press, 1902, XXXV 908 pp. Edition limited to 1000 numbered and signed copies.

^[27] Judge.

The chief credit for bringing together the materials for the First Folio, in 1623, is believed to be due to William Jaggard. Some ten years earlier he had acquired the printing-privileges of certain of the quartos. Edward Blount, whose name appears as publisher on the title page with that of Isaac Jaggard, was merely a stationer, so that the actual printing was solely under the charge of the latter, who seems, at this time, to have been entrusted with this department of the business. However, Blount's services may have been valuable since he had better literary taste than the Jaggards possessed.

In spite of certain evident faults of proportion, the portrait of Shakespeare engraved by Martin Droeshout for the title page of the 1623 Folio bears

internal evidence of being a fairly good likeness, for the face possesses a marked individuality. There is a belief that it was taken from the so-called "Flower" portrait, now in the Shakespeare Memorial Gallery at Stratford-upon-Avon, and which is conjectured to have been painted in 1609, at least during Shakespeare's lifetime, possibly by another Martin Droeshout, a Fleming, uncle of the engraver of the same name. This portrait was discovered, painted on a panel at Peckham Rye, bearing the inscription "Will Shakespeare^n, 1609". That it should be the original from which the Droeshout engraving was taken has been doubted, since it appears rather to resemble later states of the plate than earlier ones. While Ben Jonson, who had seen Shakespeare so often, may have been partly moved to bestow undue praise upon the Folio portrait, in the lines he furnished the publishers to be placed immediately facing it, by his wish to say a good word for their publication, he would scarcely have made use of such superlative terms had he not considered it to be at least a fairly good likeness. Jonson's lines have been so often printed that few are unacquainted with them, but as illustrating the above remarks they can be repeated here, in the old spelling and form of the First Folio:

TO THE READER.

This Figure, that thou here seest put,
It was for gentle Shakespeare cut;
Wherein the Graver has a strife
With Nature, to out-doo the life:
O, could he but have drawne his wit
As well in brasse, as he hath hit
His face; the Print would then surpass
All, that was ever write in brasse.
But, since he cannot, Reader, looke
Not on his Picture, but his Booke.

B.I.

A most attractive and instructive exhibition of reproductions of the portraits of Shakespeare, or supposedly of him, was shown at the rooms of the Grolier Club, April 6-29, 1916. The catalogue ^[28] embraces 436 numbers, illustrating all the principal types. The exhibition also comprised the principal editions of the poet's plays, from the First Folio of 1623 to the great Variorum Edition by Dr. Furness, begun in 1871.

[28] Catalogue of an exhibition illustrative of the text of Shakespeare's plays, as published in edited editions, together with a large collection of engraved portraits of the poet. New York, The Grolier Club, April 6-29, 1916, vi+114 pp.

For the Tercentenary of Shakespeare's birth, celebrated in April, 1864, a special commemorative medal was struck in England, designed by Mr. J. Moore. The obverse shows a profile head of the poet, in the modelling of which the artist seems to have been chiefly influenced by the Stratford bust. This fundamental type he has not unskilfully combined with that of the Droeshout print in the First Folio, the dome-like forehead being evidently suggested by the latter. The nose is more accentuated than in the bust, and the mouth, though still small, is somewhat firmer. Toward the edge of the field are disposed the titles of his various works, as though radiating from the head, and in the exergue is his signature, framed by a half-garland over which extends a mace. The tribute offered to Shakespeare by the Muses, figured on the reverse, is a rather stiff and conventional composition. [29]

[29] W. Sharp Ogden, "Shakspeare's Portraits: painted, graven, and medallie", in *The British Numismatic Journal, and Proceedings of The British Numismatic Society*, 1910, London, 1911, pp. 143-198; see p. 189.

For those who may wish to see the original form of the passages regarding precious stones in the text of the First Folio, of 1623, the page and column references have been given here. In this text the three sections into which the plays have been divided, Comedies, Histories, and Tragedies, are separately paged; moreover, the pagination offers a number of irregularities. *Troilus and Cressida*, added at the end of the "Histories", has page numbers on a couple of leaves neither connected with what precedes nor with what follows, the remainder of the pages bearing no figures; furthermore, there are several obvious, though unimportant, misprints. *Pericles*, first issued in Folio, in the Third Folio, of 1664, is therein separately paged, as are the other of the plays attributed to Shakespeare printed therein, in continuation of the series of the First and Second Folios. This play had, however, previously appeared six times in quarto in the years 1609, 1611, 1619, 1630, 1635 and 1639.

PRECIOUS STONES MENTIONED IN THE PLAYS OF SHAKESPEARE

DIAMOND

I see how thine eye would emulate the diamond.

Merry Wives of Windsor, Act iii, sc. 3, l. 59.

"Comedies", p. 58 [50], col. A, line 31.

DIAMOND

Give me the ring of mine you had at dinner,

Or, for my diamond, the chain you promised.

Comedy of Errors, Act iv, sc. 3, l. 70.

"Comedies", p. 94, col. B, lines 61, 62.

DIAMOND

Sir, I must have that diamond from you.—

There, take it.

Comedy of Errors, Act v, sc. 1, l. 391.

"Comedies", p. 99, col. B, line 58.

DIAMOND

A lady walled about with diamonds!

Love's Labour's Lost, Act v, sc. 2, l. 3.

"Comedies", p. 137, col. A, line 6.

DIAMOND

A diamond gone, cost me two thousand ducats in
Frankfort!

Merchant of Venice, Act iii, sc. 1, l. 87.

"Comedies", p. 173, col. A, line 62.

DIAMOND

Set this diamond safe
In golden palaces, as it becomes.
Henry VI, Pt. I, Act v, sc. 3, l. 169.
"Histories", p. 116, col. B, line 54.

DIAMOND

A heart it was, bound in with diamonds.
Henry VI, Pt. II, Act iii, sc. 2, l. 107.
"Histories", p. 134, col. A, line 46.

DIAMOND

Not deck'd with diamonds and Indian stones,
Nor to be seen.
Henry VI, Pt. III, Act iii, sc. 1, l. 63.
"Histories", p. 158, col. B, line 25.

DIAMOND

One day he gives us diamonds, next day stones.
Timon of Athens, Act iii, sc. 6, l. 131.
"Tragedies", p. 89, col. B, line 56.

DIAMOND

This diamond he greets your wife withal.
Macbeth, Act ii, sc. 1, l. 15.
"Tragedies", p. 136, col. A, line II.

DIAMOND

Which parted thence,
As pearls from diamonds dropp'd.
King Lear, Act iv, sc. 3, l. 24.
Omitted in First Folio.

DIAMOND

This diamond was my mother's; take it, heart;
But keep it till you woo another wife.

Cymbeline, Act i, sc. 1, l. 112.
"Tragedies", p. 370, col. A, line 45.

DIAMOND

She went before others I have seen, as that diamond
of
yours outlustres many I have beheld.

Cymbeline, Act i, sc. 4, l. 78.
"Tragedies", p. 372, col. A, line 53.

DIAMOND

I have not seen the most precious diamond that is,
nor
you the lady.

Cymbeline, Act i, sc. 4, l. 81.
"Tragedies", p. 372, col. A, line 55.

DIAMOND

I shall but lend my diamond till your return.
Cymbeline, Act. i, sc. 4, l. 153.
"Tragedies", p. 372, col. B, line 59.

DIAMOND

My ten thousand ducats are yours; so is your
diamond too.
Cymbeline, Act i, sc. 4, l. 163.
"Tragedies", p. 373, col. A, line 1.

DIAMOND

It must be married
To that your diamond.
Cymbeline, Act ii, sc. 4, l. 98.
"Tragedies", p. 389 [379], col. A, lines 42, 43.

DIAMOND

That diamond upon your finger, say,
How came it yours?

Cymbeline, Act v, sc. 5, l. 137.
"Tragedies", p. 396, col. A, line 51.

DIAMOND

To me he seems like diamond to glass.

Pericles, Act ii, sc. 3, l. 36.
Third Folio, 1664, p. 7, col. B, line 38;
separate pagination.

DIAMOND

You shall, like diamonds, sit about his crown.

Pericles, Act ii, sc. 4, l. 53.
Third Folio, 1664, p. 8, col. B, line 42.

DIAMOND

The diamonds of a most praised water
Do appear, to make the world twice rich.

Pericles, Act iii, sc. 2, l. 102.
Third Folio, 1664, p. 11, col. B, line 13.

RUBY

The impression of keen whips I'd wear as rubies.

Measure for Measure, Act ii, sc. 4, l. 101.
"Comedies", p. 69, col. B, line 63.

RUBY

Her nose, all o'er embellished with rubies,
carbuncles,
sapphires.

Comedy of Errors, Act iii, sc. 2, l. 138.
"Comedies", p. 92, col. A, line 49.

RUBY

Those be rubies, fairy favors.

Midsummer Night's Dream, Act ii, sc. 1, l. 12.

"Comedies", p. 148, col. A, line 35.

RUBY

Over thy wounds now do I prophesy,—Which,
like dumb mouths, do ope their ruby lips.

Julius Caësar, Act iii, sc. 1, l. 260.

"Tragedies", p. 120, col. B, lines 34, 35.

RUBY

And keep the natural ruby of your cheeks,
When mine is blanch'd with fear.

Macbeth, Act iii, sc. 4, l. 115.

"Tragedies", p. 142, col. B, line 17.

RUBY

But kiss; one kiss! Rubies unparagon'd,
How dearly they do't!

Cymbeline, Act ii, sc. 2, l. 17.

"Tragedies", p. 376, col. B, line 18.

SAPPHIRE

Like sapphire, pearl and rich embroidery.

Merry Wives of Windsor, Act v, sc. 5, l. 75.

"Comedies", p. 51, col. A, line 66 (last).

SAPPHIRE

Her nose, all o'er embellished with rubies,
carbuncles,
sapphires.

Comedy of Errors, Act iii, sc. 2, l. 138.

"Comedies", p. 92, col. A, line 49.

CHRYSOLEITE

If heaven would make me such another world
Of one entire and perfect chrysolite.

Othello, Act v, sc. 2, l. 145.
"Tragedies", p. 337, col. A, line 5.

TURQUOISE

It was my turquoise; I had it of Leah when I was a
bachelor.

Merchant of Venice, Act iii, sc. 1, l. 126.
"Comedies", p. 173, col. B, line 32.

OPAL

For thy mind is a very opal.

Twelfth Night, Act ii, sc. 4, l. 77.
"Comedies", p. 262, col. B, line 45.

AGATE (CAMEO)

An agate very vilely cut.

Much Ado About Nothing, Act iii, sc. 1, l. 65.
"Comedies", p. 110, col. A, line 25.

AGATE (CAMEO)

His heart like an agate with your print impress'd.

Love's Labour's Lost, Act ii, sc. 1, l. 236.
"Comedies", p. 127, col. B, line 62 (last).

AGATE (CAMEO)

I was never manned with an agate till now.

II Henry IV, Act i, sc. 2, l. 19.
"Histories", p. 76, col. B, line 10.

AGATE (CAMEO)

Agate-ring, pirke-stocking, caddis-garter, smooth-
tongue.

I Henry IV, Act ii, sc. 4, l. 78.

"Histories", p. 56, col. A, line 53.

AGATE (CAMEO)

In shape no bigger than an agate-stone
On the forefinger of an alderman.

Romeo and Juliet, Act i, sc. 4, l. 55.
"Tragedies", p. 57, col. A, lines 20, 21.

AMBER

Her amber hair for foul hath amber quoted.
Love's Labour's Lost, Act iv, sc. 3, l. 87.
"Comedies", p. 133, col. A, line 52.

AMBER

With amber bracelets, beads, and all this knavery.
Taming of the Shrew, Act iv, sc. 3, l. 58.
"Comedies", p. 223, col. B, line 62.

AMBER

Their eyes purging thick amber and plum-tree gum.
Hamlet, Act ii, sc. 2, l. 201.
"Tragedies", p. 261, col. B, line 42.

CORAL

Of his bones are coral made.
The Tempest, Act i, sc. 2, l. 397.
"Comedies", p. 5, col. A, line 54.

CORAL

I saw her coral lips to move.
Taming of the Shrew, Act i, sc. 1, l. 179.
"Comedies", p. 211, col. B, line 57.

JET

There is more difference between thy flesh and hers
than
between jet and ivory.

Merchant of Venice, Act iii, sc. 1, l. 42.

"Comedies", p. 173, col. A, line 18.

JET

What color is my gown of?—Black, forsooth: coal-
black
as jet.

II Henry VI, Act ii, sc. 1, l. 112.

"Histories", p. 126, col. B, line 61.

JET

Two proper palfreys, black as jet,
To hale thy vengeful waggon swift away.

Titus Andronicus, Act v, sc. 2, l. 50.

"Tragedies", p. 49, col. B, line 7.

CARBUNCLE

Her nose, all o'er embellished with rubies,
carbuncles,
sapphires.

Comedy of Errors, Act iii, sc. 2, l. 138.

"Comedies", p. 92, col. A, line 49.

CARBUNCLE

A carbuncle entire, as big as thou art,
Were not so rich a jewel.

Coriolanus, Act i, sc. 4, l. 55.

"Tragedies", p. 5, col. B, line 7.

CARBUNCLES

O'er sized with coagulate gore,
With eyes like carbuncles.

Hamlet, Act ii, sc. ii, l. 485.

"Tragedies", p. 263, col. B, line 50.

CARBUNCLE

Were it carbuncled
Like holy Phœbus' car.
Antony and Cleopatra, Act iv, sc. 8, l. 28.
"Tragedies", p. 360, col. B, line 57.

CARBUNCLE

Had it been a carbuncle
Of Phœbus' wheel.
Cymbeline, Act v, sc. 5, l. 189.
"Tragedies", p. 396, col. B, line 41.

EMERALD

In emerald tufts, flowers purple, blue, and white.
Merry Wives of Windsor, Act v, sc. 5, l. 74.
"Comedies", p. 51, col. A, line 65.

PEARLS

Full fathom five thy father lies;
Of his bones are coral made;
Those are pearls that were his eyes.
Tempest, Act i, sc. 2, l. 398.
"Comedies", p. 5, col. A, lines 51-33.

PEARLS

She is mine own,
And I as rich in having such a jewel
As twenty seas, if all their sand were pearl.
Two Gentlemen of Verona, Act ii, sc. 4, l. 170.
"Comedies", p. 26, col. B, lines 34-36.

PEARLS

A sea of melting pearl, which some call tears.

Two Gentlemen of Verona, Act iii, sc. 1, l. 224.
"Comedies", p. 30, col. B, line 2.

PEARLS

But pearls are fair; and the old saying is,
Black men are pearls in beauteous ladies' eyes
'Tis true; such pearls as put out ladies' eyes.
Two Gentlemen of Verona, Act v, sc. 2, l. 11.
"Comedies", p. 36, col. B, lines 10-12.

PEARLS

Like sapphire, pearl and rich embroidery
Buckled below fair knighthood's bending knee.
Merry Wives of Windsor, Act v, sc. 5, l. 75.
"Comedies", p. 51, col. A, lines 65, 66 (last).

PEARLS

Laced with silver, set with pearls
Much Ado About Nothing, Act iii, sc. 4, l. 20.
"Comedies", p. 112, col. B, line 65.

PEARLS

Fire enough for a flint, pearl enough for a swine.
Love's Labour's Lost, Act iv, sc. 2, l. 91.
"Comedies", p. 132, col. A, line 11.

This and these pearls to me sent Longaville.
Love's Labour's Lost, Act v, sc. 2, l. 53.
"Comedies", p. 137, col. A, line 59.

PEARLS

Will you have me, or your pearl again?
Neither of either.
Love's Labour's Lost, Act v, sc. 2, l. 458.
Comedies", p. 140, col. B, line 58.

PEARLS

Decking with liquid pearl the bladed grass.

Midsummer Night's Dream, Act i, sc. 1, l. 211.

"Comedies", p. 147, col. A, line 6.

PEARLS

I must go seek some dewdrops here

And hang a pearl in every cowslip's ear.

Midsummer Night's Dream, Act ii, sc. 1, l. 15.

"Comedies", p. 148, col. A, line 38.

PEARLS

That same dew, which sometime in the buds

Was wont to swell like round and orient pearls.

Midsummer Night's Dream, Act iv, sc. 1, l. 57.

"Comedies", p. 157, col. B, lines 9, 10.

PEARLS

Rich honesty dwells like a miser, sir, in a poor house;

as

your pearl in your foul oyster.

As You Like It, Act v, sc. 4, l. 63.

"Comedies", p. 206, col. A, line 12.

PEARLS

Their harness studded all with gold and pearl.

Taming of the Shrew, Introd., sc. 2, l. 44.

"Comedies", p. 209, col. B, line 33.

PEARLS

Fine linen, Turkey cushions boss'd with pearls

Valance of Venice gold.

Taming of the Shrew, Act ii, sc. 1, l. 355.

"Comedies", p. 217, col. B, line[Pg 093] 32.

PEARLS

Why, sir, what 'cerns it you if I wear pearl and gold?
Taming of the Shrew, Act v, sc. 1, l. 77.
"Comedies", p. 227, col A, line 22.

PEARLS

This pearl she gave me, I do feel't and see't.
Twelfth Night, Act iv, sc. 3, l. 2.
"Comedies", p. 271, col. B, line 61.

PEARLS

Draws those heaven-moving pearls from his poor
eyes.
King John, Act ii, sc. 1, l. 169.
"Histories", p. 4, col. B, line 55.

PEARLS

Our chains and our jewels.—
Your brooches, pearls and ouches.
II Henry IV, Act ii, sc. 4, l. 53.
"Histories", p. 82, col. B, line 28.

PEARLS

The crown imperial,
The intertissued robe of gold and pearl.
Henry V, Act iv, sc. 1, l. 279.
"Histories", p. 85 (bis, number repeated), col. B,
line 13.

PEARLS

Wedges of gold, great anchors, heaps of pearl,
Inestimable stones, unvalued jewels.
Richard III, Act i, sc. 4, l. 26.
"Histories", p. 180, col. A, line 12.

PEARLS

The liquid drops of tears that you have shed
Shall come again, transform'd to orient pearl.

Richard III, Act iv, sc. 4, l. 322.
"Histories", p. 198, col. A, lines 16, 17.

PEARLS

Her bed is India; there she lies, a pearl.

Troilus and Cressida, Act i, sc. 1, l. 103.
"Histories", page irregularly numbered 79,
col. A, line 8. P. 589 of facsimile.

PEARLS

She is a pearl
Whose price hath launch'd above a thousand ships.

Troilus and Cressida, Act ii, sc. 2, l. 81.
Unnumbered page, 596 of facsimile, col. A, line 19.

PEARLS

I will be bright, and shine in pearl and gold.

Titus Andronicus, Act ii, sc. 1, l. 19.
"Tragedies", p. 35, col. B, line 30.

PEARLS

This is the pearl that pleased your empress' eye.

Titus Andronicus, Act v, sc. 1, l. 42.
"Tragedies", p. 48, col. A, line 21.

PEARLS

I see thee compass'd with thy kingdom's pearl.

Macbeth, Act v, sc. 8, l. 56.
"Tragedies", p. 151, col. B, line 32.

PEARLS

Hamlet, this pearl is thine.

Hamlet, Act v, sc. 2, l. 293.
"Tragedies", p. 281, col. A, line 15.

PEARLS

What guests were in her eyes; which parted thence,
As pearls from diamonds dropp'd.

Lear, Act iv, sc. 3, l. 24.
Omitted in First Folio.

PEARLS

Like the base Indian, ^[30] threw a pearl away
Richer than all his tribe.

Othello, Act v, sc. 2, l. 347.
"Tragedies", p. 338, col. B, line 53.

^[30] "Iudean" in text.

PEARLS

He kiss'd,—the last of many doubled kisses,—
This orient pearl.

Antony and Cleopatra, Act i, sc. 5, l. 41.
"Tragedies", p. 344, col. B, lines 22, 23.

PEARLS

I'll set thee in a shower of gold, and hail
Rich pearls upon thee.

Antony and Cleopatra, Act ii, sc. 5, l. 46.
"Tragedies", p. 348, col. B, lines 10, 11.

ROCK-CRYSTAL

His mistress
Did hold his eyes lock'd in her crystal looks.

Two Gentlemen of Verona, Act ii, sc. 4, l. 89.
"Comedies", p. 26, col. A, line 17.

ROCK-CRYSTAL

Methough all his senses were lock'd in his eye
As jewels in crystal for some prince to buy.
Love's Labour's Lost, Act ii, sc. 1, l. 243.
"Comedies", p. 128, col. A, lines 6, 7.

ROCK-CRYSTAL

One, her hairs were gold, crystal the other's eyes.
Idem, Act iv, sc. 3, l. 142.
"Comedies", p. 133, line 46.

ROCK-CRYSTAL

To what, my love, shall I compare thine eye?
Crystal is muddy.
Midsummer Night's Dream, Act iii, sc. 2, l. 139.
"Comedies", p. 154, col. A, line 54.

ROCK-CRYSTAL

With these crystal beads heaven shall be bribed
To do him justice.
King John, Act ii, sc. 1, l. 171.
"Histories", p. 4, col. B, lines 57, 58.

ROCK-CRYSTAL

The more fair and crystal is the sky,
The uglier seem the clouds that in it fly.
Richard II, Act i, sc. 1, l. 41.
"Histories", p. 23, col. A, line 41 (last).

ROCK-CRYSTAL

Go, clear thy crystals.
Henry V, Act ii, sc. 3, l. 56.
"Histories", p. 75, col. B, line 65.

ROCK-CRYSTAL

Comets, importing change of times and states,

Brandish your crystal tresses in the sky.

I Henry VI, Act i, sc. 1, l. 3.

"Histories", p. 96, col. A, lines 2, 3.

ROCK-CRYSTAL

But in that crystal scales let there be weigh'd

Your lady's love against some other maid.

Romeo and Juliet, Act i, sc. 2, l. 101.

"Tragedies", p. 55, col. B, lines 51, 52.

ROCK-CRYSTAL

Thy crystal window ope; look out.

Cymbeline, Act v, sc. 4, l. 81.

"Tragedies", p. 394, col. A, line 12.

The following table is arranged according to the frequency of precious stone mentions. [\[31\]](#)

The plays rank as follows:

First published	Probably written		
1623	1609	<i>Cymbeline</i>	10 (diamond 7, ruby 1, carbuncle 1, rock-crystal 1).
1598.	1591.	<i>Love's Labour's Lost</i>	8 (pearl 3, rock-crystal 2, diamond 1, amber 1, agate 1).
1600	1597	<i>Merry Wives Of Windsor</i>	5 (pearl 1, diamond 2, emerald 1, sapphire 1).
1623	1591.	<i>Comedy of Errors</i>	5 (diamond 2, ruby 1, sapphire 1, carbuncle 1)

1600.	1595.	<i>Midsummer Night's Dream</i>	5 (pearl 3, ruby 1, rock-crystal 1).
1623.	1596.	<i>Taming of the Shrew</i>	5 (pearl 3, amber 1, coral 1).
1623.	1591.	<i>Two gentlemen of Verona</i>	4 (pearl 3, rock-crystal 1).
1594.	1593.	<i>Titus Andronicus</i>	3 (pearl 2, jet 1).
1603	1602	<i>Hamlet</i>	3 (pearl, amber, carbuncle).
1623	1606	<i>Macbeth</i>	3 (diamond, ruby, pearl).
1609	1607	<i>Pericles</i>	3 (all diamond).
1623	1608	<i>Antony and Cleopatra</i>	3 (pearl 2, carbuncle 1).
1597	1591	<i>Romeo and Juliet</i>	2 (rock-crystal, agate).
1623	1592	<i>I Henry VI</i>	2 (diamond and rock-crystal).
1623	1592	<i>II Henry VI</i>	2 (diamond and jet).
1597	1592-3	<i>Richard III</i>	2 (both pearl).
1600	1594	<i>Merchant of Venice</i>	2 (turquoise, jet).
1623	1594	<i>King John</i>	2 (pearl, rock-crystal).

1623	1597	<i>II Henry IV</i>	2 (pearl, agate).
1600	1598	<i>Henry V</i>	2 (pearl, crystal).
1600	1599	<i>Much Ado About Nothing</i>	2 (pearl, agate).
1623	1599	<i>Twelfth Night</i>	2 (pearl, opal).
1609	1603	<i>Troilus and Cressida</i>	2 (both pearl).
1622	1604	<i>Othello</i>	2 (pearl, chrysolite).
1608	1606	<i>Lear</i>	2 (pearl, diamond).
1623	1611	<i>Tempest</i>	2 (pearl, coral).
1623	1592	<i>III Henry VI</i>	1 (diamond).
1597	1593	<i>Richard II</i>	1 (rock-crystal).
1598	1597	<i>I Henry IV</i>	1 (agate).
1623	1599	<i>As You Like It</i>	1 (pearl).
1623	1601	<i>Julius Cæsar</i>	1 (ruby).
1623	1604	<i>Measure for Measure</i>	1 (ruby).
1623	1607	<i>Timon of Athens</i>	1 (diamond).

1623

1608

Coriolanus

1 (carbuncle).

[31] Data of first publication contributed by Miss Henrietta C. Bartlett.

PRECIOUS STONES MENTIONED IN POEMS OF SHAKESPEARE.

DIAMOND

The diamond—why 'twas beautiful and hard.
"Lover's Complaint", l. 211.

SAPPHIRE

The heaven-hued sapphire and the opal blend
With objects manifold.
Idem, l. 215.

PEARLS

Her tears began to turn their tide,
Being prison'd in her eye like pearls in glass.
"Venus and Adonis", l. 980.
G, verso, l. 1, 2.

PEARLS

And wiped the brinish pearl from her bright eyes.
"Lucrece", l. 1213.
I 2, l. 2.

PEARLS

Those round clear pearls of his, that move thy pity,
Are balls of quenchless fire to burn thy city.
Idem, l. 1553.
L. 2, verso, l. 6, 7.

PEARLS

Of paled pearls and rubies red as blood.

"Lover's Complaint", l. 198.

PEARLS

Ah! but those tears are pearls which thy love sheds.

Sonnet XXXIV, l. 13.

C 4, l. 13.

PEARLS

Bright orient pearl, alack, too timely shaded!

"Passionate Pilgrim", l. 133.

B 4, l. 3.

OPAL

The heaven-hued sapphire and the opal blend
With objects manifold.

"Lover's Complaint", l. 215.

RUBY

Once more the ruby-colour'd portal open'd.

"Venus and Adonis", l. 451.

D ii, verso, l. 1.

RUBY

Of paled pearls and rubies red as blood.

"Lover's Complaint", l. 198.

EMERALD

The deep-green emerald, in whose fresh regard
Weak sights their sickly radiance do amend.

Idem, l. 213.

ROCK-CRYSTAL

But hers through which the crystal tears gave light,
Shone like the moon in water seen by night.

"Venus and Adonis", l. 491.

D iii, l. 16, 17.

ROCK-CRYSTAL

Nor thy soft hands, sweet lips, and crystal eyne.

"Venus and Adonis", l. 633.

E ii, l. 15.

ROCK-CRYSTAL

The crystal tide that from her two cheeks fair
In the sweet channel of her bosom dropt.

Idem, l. 957.

G, l. 3, 4.

ROCK-CRYSTAL

Her eyes seen in the tears, tears in her eye;
Both crystals, where they view'd each other's sorrow.

Idem, l. 962, 963.

G, l. 8, 9.

ROCK-CRYSTALS

Through crystal walls each little mote will peep.

"Lucrece", l. 1251.

I 2, verso, l. 19.

ROCK-CRYSTAL

A closet never pierced with crystal eyes.

Sonnet XLVI, l. 6.

D 2, verso, l. 6.

ROCK-CRYSTAL

Favours from a maund ^[32] she drew
Of amber, crystal, and of beaded jet.

"Lover's Complaint", l. 37.

[32] Basket, or hamper.

ROCK-CRYSTAL

Who glazed with crystal gate the glowing roses.
"Lover's Complaint", l. 286.

AMBER

With coral clasps and amber studs.
"Passionate Pilgrim", l. 366.
D 4, verso, l. 2.

AMBER

Favours from a maund she drew
Of amber, crystal, and of beaded jet.
"Lover's Complaint", l. 37.

JET

as above.

CORAL

That sweet coral mouth
Whose precious taste her thirsty lips well knew.
"Venus and Adonis", l. 542.
D iv, l. 20, 21.

CORAL

Her alabaster skin,
Her coral lips, her snow white dimpled chin.
"Lucrece", l. 420.
D 3, l. 7.

CORAL

Like ivory conduits coral cisterns filling.
Idem, l. 1234.

I 2, verso, l. 2.

CORAL

Coral is far more red than her lips' red.

Sonnet CXXX, l. 2.

H 4, l. 2.

CORAL

A belt of straw and ivy buds.

With coral clasps and amber studs.

"Passionate Pilgrim", l. 366.

D 4, verso, l. 1, 2. [\[33\]](#)

[\[33\]](#) References are here given to the original editions of "Venus and Adonis", 1593 (unique copy in the Malone Collection in the Bodleian Library, Oxford); "Lucrece", 1594; "Passionate Pilgrim", 1599, and Sonnets, 1609. As there is no continuous pagination, the letters and numbers refer to the page signatures and to the line of the page.

While it cannot be regarded as certain that whenever Shakespeare writes of jewels or of rings he means those in which precious stones were set, several of the passages more or less clearly indicate this, and we therefore present here the more characteristic of the lines in question:

A Death's face in a ring.

Love's Labour's Lost, Act v, sc. 2, l. 616.

"Comedies", p. 142, col. A, line 36.

The dearest ring in Venice will I give you.

Merchant of Venice, Act iv, sc. 1, l. 435.

"Comedies", p. 181, col. B, line 27.

Diana. O behold this ring
Whose high respect and rich validity
Did lack a parallel; yet for all that
He gave it to a commoner of the camp,
If I be one.

Count. He blushes, and 'tis it:
Of six preceding ancestors, that gem,

Conferr'd by testament to the sequent issue,
Hath it been owned and worn.

All's Well That Ends Well, Act v, sc. 3, l. 191-198.
"Comedies", p. 253, col. A, lines 1-8.

My daughter! O my ducats! O my daughter!
Fled with a Christian! O my Christian ducats!
Justice! the law! my ducats and my daughter!
A sealed bag, two sealed bags of ducats,
Of double ducats, stolen from me by my daughter!
And jewels, two stones, two rich and precious stones,
Stolen by my daughter! Justice! find the girl;
She hath the stones upon her, and the ducats.

Merchant of Venice, Act ii, sc. 8, l. 15-22.
"Comedies", p. 171, col. B, lines 23-30.

I would my daughter were dead at my foot, and the
jewels in her ear!

Merchant of Venice, Act iii, sc. 1, l. 92.
"Comedies", p. 173, col. B, lines 1, 2.

Sweet are the uses of adversity,
Which, like the toad, ugly and venomous,
Wears yet a precious jewel in his head.

As You Like It, Act ii, sc. 1, l. 13-15.
"Comedies", p. 190, col. A, lines 10-12.

Win her with gifts, if she respect not words:
Dumb jewels often in their silent kind
More than quick words do move a woman's mind.

Two Gentlemen of Verona, Act iii, sc. 1, l. 89-91.
"Comedies", p. 29, col. A, lines 63-65.

I frown the while; and perchance wind up my watch,
or play with my—some rich jewel.

Twelfth Night, Act ii, sc. 5, l. 64-66.
"Comedies", p. 263, col. B, lines 32, 33.

A jewel in a ten-times-barr'd-up chest

Is a bold spirit in a loyal breast.

King Richard II, Act i, sc. 1, l. 180, 181.
"Histories", p. 24, col. B, lines 28, 29.

This royal throne of Kings, this scepter'd isle,
This earth of majesty, this seat of Mars,
This other Eden, demi-paradise,
This fortress built by Nature for herself
Against infection and the hand of war,
This happy breed of men, this little world,
This precious stone set in the silver sea,
Which serves it in the office of a wall
Or as a moat defensive to a house,
Against the envy of less happier lands,
This blessed plot, this earth, this realm, this England.

King Richard II, Act ii, sc. 1, l. 40-46.
"Histories", p. 28, col. B, lines 17-23.

In argument and proof of which contract,
Bear her this jewel, pledge of my affection.

I Henry VI, Act v, sc. 2, l. 46, 47.
"Histories", p. 115, col. A, lines 8, 9.

It seems she hangs upon the cheek of night,
Like a rich jewel in an Ethiop's ear;
Beauty too rich for use, for earth too dear.

Romeo and Juliet, Act i, sc. 5, l. 47-49.
"Tragedies", p. 57, col. B, lines 59-61.

But chiefly to take thence from her dead finger
A precious ring, a ring that I must use
In dear employment.

Romeo and Juliet, Act v, sc. 3, l. 30-32.
"Tragedies", p. 75, col. A, lines 34-36.

A striking proof that Shakespeare had no fear of tautology when he wished to strengthen the impression of a word by constant reiteration is given in the Merchant of Venice (Act v, sc. 2), whence we have already quoted a few

lines. The passage concerns the disposal by Bassanio of a ring he had received from Portia, and he answers her thus in the First Folio text: ^[34]

Bassanio. Sweet Portia,
If you did know to whom I gave the Ring,
If you did know for whom I gave the Ring,
And would conceive for what I gave the Ring,
And how unwillingly I left the Ring,
When naught would be accepted but the Ring,
You would abate the strength of your displeasure.

Portia.
If you had knowne the virtue of the Ring,
Or halfe her worthinesse that gave the Ring,
Or your owne honour to contains the Ring,
You would not then have parted with the Ring.

[34] First Folio, "Comedies", p. 183, col. B, lines 36-46.

It was probably more than a coincidence that Shakespeare's first printed book, "Venus and Adonis", was published, in 1593, by a fellow-townsmen, Richard Field, who had come up to London from Stratford when a mere boy. Undoubtedly, when Shakespeare met him in the bustle of city life, the common memories of their quieter native town served at once as an introduction and as a link between them. Field also published Shakespeare's "Lucrece" in the year 1594. He had been a freeman of the Stationers' Company from February 6, 1587, and died either in the year the First Folio was issued, or in the succeeding year, 1624.



Printer's mark of Richard Field, as shown on the title-page of the first edition of Shakespeare's "Venus and Adonis", 1593, the unique copy of which is in the Bodleian Library, Oxford. A hand emerging from a cloud upholds the "Anchor of Hope", about which are twined two laurel branches.

*** END OF THE PROJECT GUTENBERG EBOOK SHAKESPEARE
AND PRECIOUS STONES ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.